

Honeywell

Experion and TPS Systems

**T5400-based Honeywell Workstation
Planning, Installation, and Service Guide**

HW10-100

EP-DPCXX9

Release Independent

12/2008

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About This Document

This document contains installation and service information for the Dell Precision T5400 Workstation (T5400-based Honeywell workstation). The instructions and service information contained herein address the workstation itself, and assumes that associated network communication equipment has been pre-installed by the Honeywell factory or has manuals dedicated to its installation and service. This workstation is not a standard Dell model and cannot be ordered independently from Dell.

Release Information

Document Name	Document ID	Release Number	Publication Date
T5400-based Honeywell Workstation Planning, Installation, and Service Guide – HW10	EP-DPCXX9	Release Independent	Initial Issue

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title
TPS Users
<i>TPS System Site Planning</i>
<i>TPS System Implementation Guide for Windows 2003/XP</i>
<i>TPS System Planning Guide for Windows 2003/XP</i>
<i>TPS System Administration Guide for Windows XP/2003</i>
Experion PKS Users
<i>Experion PKS Overview</i>
<i>Experion PKS Software Installation and Upgrade Guide</i>
<i>Server and Client Planning Guide</i>
<i>Server and Client Configuration Guide (for Experion PKS)</i>
<i>Experion PKS Operators Guide</i>
FTE Users
<i>Fault Tolerant Ethernet Installation and Service Guide</i>
Console Documents
<i>Honeywell Icon Series Console Planning, Installation, and Service Guide</i>
12/2008 T5400-based Honeywell Workstation Planning, Installation, and Service Guide iii
Release Independent Honeywell

Document Title

Universal Station Service

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	REFERENCE - INTERNAL: Identifies an additional source of information within the book set.
	CAUTION: Indicates a potentially hazardous situation, which if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation, which if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING, Risk of electrical shock: Potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.
	ESD HAZARD: Danger of an electro-static discharge to which equipment may be sensitive. Observe precautions for handling electrostatic sensitive devices.

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1. System Planning

1.1 Introduction

Platform overview

Platforms sold by Honeywell are engineered for the process control mission of Experion and TPS systems to provide consistent, robust performance. Through an extensive qualification process, Honeywell defines specific peripheral devices, slot locations, and BIOS settings for best performance and reliability, sometimes even adding cooling fans for longer service. Honeywell platforms are then built to Honeywell specifications by the computer manufacturer.

Honeywell engineering has tested the T5400-based Honeywell workstation with other Honeywell hardware and software and has qualified its use for specific configurations as identified in the Software Change Notice (SCN). This workstation is not a standard Dell model and cannot be ordered independently from Dell. The Technical Assistance Center (TAC) is trained on and fully supports Honeywell platforms. Use of any other workstation, including a similar Dell model, is considered a project special and its TAC support is limited according to the services policy.

The workstation platform provides computer-based functionality for the Experion and TPS system, including Global User Stations (GUS) and APP Nodes. For TPS systems, the platforms have an LCNP4M card installed allowing connection to the TotalPlant Process Network (TPN) coax through the LCN Media Access Unit (MAU). Additionally, these platforms connect to the Ethernet through a standard Ethernet adapter card or the optional FTE Ethernet adapter card.

Furniture options

The workstation may be mounted in a variety of Honeywell furniture. The following tables lists the types of furniture in which the workstation can be mounted and indicates whether it uses a fixed rail mount or slide tray mount.

Furniture Type	Mounting Option	
	Fixed (rails)	Slide (tray)
Rittal Cabinet	Yes	Yes
Classic Console	Yes	Yes
Z Console	Yes	No
EZ Console	Yes	No
Icon Series Console	N/A	N/A

1.1. Introduction

These platforms have been qualified with Microsoft Windows XP 32-bit Professional Service Pack 3 Operating System with Honeywell software. Refer to the latest SCN for software applications that have been qualified for use on the T5400-based Honeywell workstation platform.

The workstation platform must have BIOS version A04 or later.

1.2 Workstation Overview

Front view

The following figure shows the front view of the workstation. Refer to Table 1-1 for a complete description of the callouts.

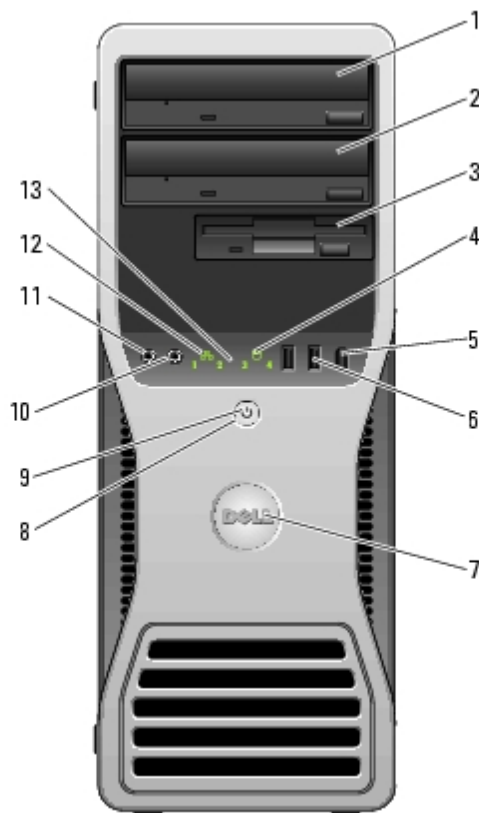


Figure 1-1 Front View of T5400-based Honeywell Workstation

Table 1-1 T5400-based Honeywell Workstation Front Panel Descriptions

No	Item	Description
1	Upper 5.25-inch drive bay	Holds a DVD-RW drive.
2	Lower 5.25inch drive bay	You can use this bay for an optional DVD-RW drive.
3	Empty Bay	
4	Hard-drive activity light	The hard drive light is on when the workstation reads data from or writes data to the hard drive. The light might also be on when a device such as your CD player is operating.
5	IEEE 1394 connector (optional)	Use the optional IEEE 1394 connector for high-speed data devices such as digital video cameras and external storage devices.
6	USB 2.0 connectors (2)	Use the front USB connectors for devices that you connect occasionally, such as flash memory keys or cameras, or for bootable USB devices (see System Setup for more information on booting to a USB device). It is recommended that you use the back USB connectors for devices that typically remain connected, such as printers and keyboards.
7	Dell™ rotatable badge	To rotate the Dell badge for tower-to-desktop conversion: remove the front panel (see Removing the Front Panel), turn it over, and rotate the plastic handle behind the badge.
8	Power button	Press to turn on the workstation. NOTICE: To avoid losing data, do not use the power button to turn off the workstation. Instead, perform an operating system shutdown. NOTE: The power button can also be used to wake the workstation or to place it into a power-saving state. See Power Management for more information.

No	Item	Description
9	Power light	The power light illuminates and blinks or remains solid to indicate different states: - o No light — The workstation is turned off or in a hibernation mode. - o Steady green — The workstation is in a normal operating state. - o Blinking green — The workstation is in a power-saving state. - o Blinking or solid amber — See Power Problems. To exit from a power-saving state, press the power button or use the keyboard or the mouse if it is configured as a wake device in the Windows Device Manager. For more information about sleep states and exiting from a power-saving state, see Power Management. See Diagnostic Lights for a description of light codes that can help you troubleshoot problems with your workstation.
10	Headphone connector	Use the headphone connector to attach headphones.
11	Microphone connector	Use the microphone connector to attach a personal workstation microphone for voice or musical input into a sound or telephony program.
12	Network link light	The network link light is on when a good connection exists between a 10-Mbps, 100-Mbps, or 1000-Mbps (or 1-Gbps) network and the workstation.
13	Diagnostic lights (4)	Use these lights to help you troubleshoot a workstation problem based on the diagnostic code. For more information, see Diagnostic Lights .

1. System Planning

1.2. Workstation Overview

Rear view

The following figure shows the rear view of the workstation. Refer to Table 1-2 for a complete description of the callouts. Your configuration may not have all the cards installed shown below.

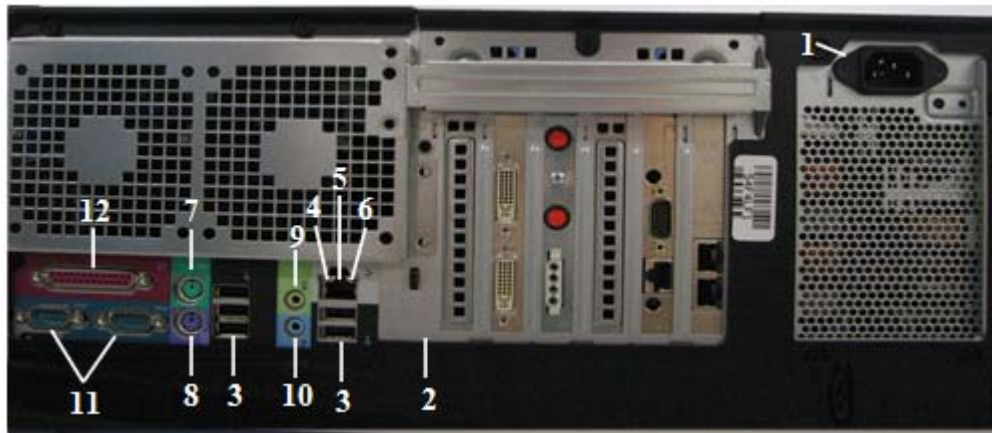


Figure 1-2 Rear View of T5400-based Honeywell Workstation

Table 1-2 T5400-based Honeywell Workstation Rear View Descriptions

No.	Connector	Description
1	Power connector	Power cable.
2	Card slots	Access connectors for any installed cards. Note: The center four slots support full length cards; one PCI-X slot, two PCI Express x16 slot, and one PCI 5V slot; and the top and bottom slots support half length cards; one PCI-X slot and one PCI Express x8 slot (wired as x4).

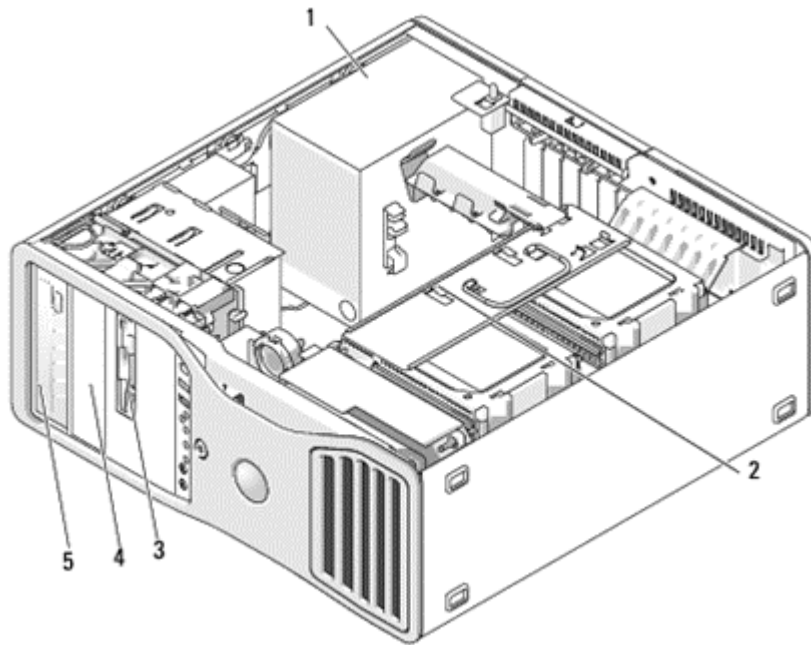
No.	Connector	Description
3	USB 2.0 connectors (5)	It is recommended that you use the front USB connectors for devices that you connect occasionally, such as flash memory keys or cameras, or for bootable USB devices. Use the back USB connectors for devices that typically remain connected, such as printers and keyboards.
4	Link integrity light	Green — A good connection exists between a 10-Mbps network and the workstation. Orange — A good connection exists between a 100-Mbps network and the workstation. Yellow — A good connection exists between a 1000-Mbps (or 1-Gbps) network and the workstation. Off — The workstation is not detecting a physical connection to the network.
5	Network adapter connector	To attach your workstation to a network or broadband device, connect one end of a network cable to either a network jack or your network or broadband device. Connect the other end of the network cable to the network adapter connector on your workstation. A click indicates that the network cable has been securely attached. Note: Do not plug a telephone cable into the network connector. On workstations with an additional network connector card, use the connectors on the card and on the back of the workstation when setting up multiple network connections (such as a separate intra and extranet). It is recommended that you use Category 5 wiring and connectors for your network. If you must use Category 3 wiring, force the network speed to 10 Mbps to ensure reliable operation.
6	Network activity light	Flashes a yellow light when the workstation is transmitting or receiving network data. A high volume of network traffic may make this light appear to be in a steady "on" state.

1. System Planning

1.2. Workstation Overview

No.	Connector	Description
7	PS/2 mouse connector	Plug a standard PS/2 mouse into the green mouse connector. Turn off the workstation and any attached devices before you connect a mouse to the workstation. If you have a USB mouse, plug it into a USB connector.
8	PS/2 keyboard connector	If you have a standard PS/2 keyboard, plug it into the purple keyboard connector. If you have a USB keyboard, plug it into a USB connector.
9	Line-out connector	Use the green line-out connector to attach most speakers with integrated amplifiers. On workstations with a sound card, use the connector on the card.
10	Line-in connector	Use the blue line-in connector to attach a playback device such as a MP3 player, CD player, or VCR. On workstations with a sound card, use the connector on the card.
11	Serial connectors (2)	Connect a serial device, such as a handheld device, to the serial port. If necessary, the address for this port can be modified through the System Setup .
12	Parallel connector	Connect a parallel device, such as a printer, to the parallel connector. If you have a USB printer, plug it into a USB connector. Note: The integrated parallel connector is automatically disabled if the workstation detects an installed card containing a parallel connector configured to the same address. For more information, see System Setup Options .

Inside enclosure

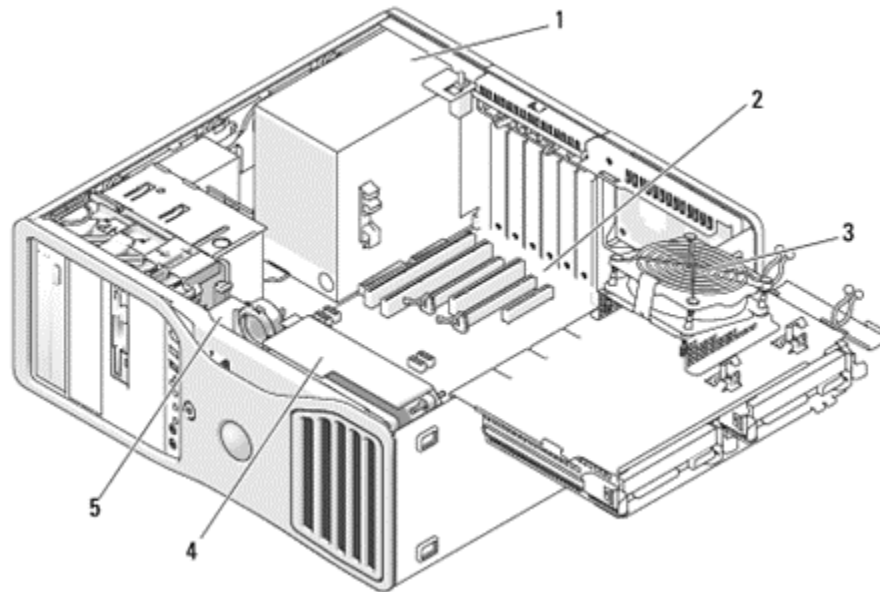


- 1 Power supply
- 2 Rotatable Hard Drive Bay
- 3 FlexBay
- 4 Lower 5.25-inch Drive Bay
- 5 Upper 5.25-inch Drive Bay

**Figure 1-3 T5400-based Workstation Enclosure
Inside View**

Inside enclosure with hard drive bay rotated out

The following picture shows the inside of the workstation enclosure and identifies the areas within the enclosure.

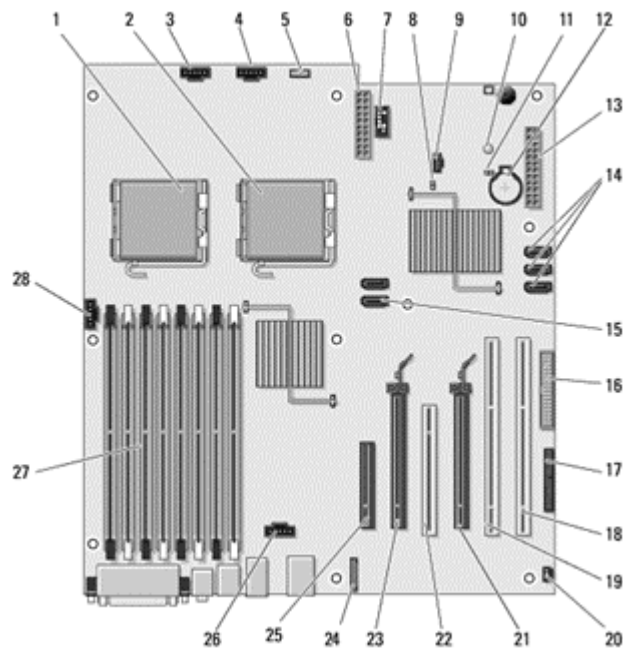


- 1 Power supply
- 2 System board (motherboard)
- 3 Memory fan on rotatable hard drive carrier
- 4 Front fan
- 5 Card fan

**Figure 1-4 T5400-based Workstation Enclosure
(Hard Drive Bay Rotated Out)**

System board

The following picture shows the system board components.



- | | |
|---|---|
| 1 primary processor connector (CPU_0) | 2 secondary processor connector (CPU_1) |
| 3 front fan connector (FAN_FRONT) | 4 card cage fan (FAN_CCAG) |
| 5 internal speaker connector (INT_SPKR) | 6 power connector (POWER2) |
| 7 USB (INT_USB) | 8 password jumper (PSWD) |
| 9 auxiliary hard-drive LED connector (AUX_LED) | 10 auxiliary power LED (AUX_PWR) |
| 11 RTC reset jumper (RTCRST) | 12 battery socket (BATTERY) |
| 13 main power connector (POWER1) | 14 SATA connectors (SATA_2, SATA_3, and SATA_4) |
| 15 SATA connectors (SATA_0, SATA_1) | 16 floppy drive (DSKT) |
| 17 front panel connector (FRONTPANEL) | 18 PCI-X card slot (SLOT6_PCIX) |
| 19 PCI-X card slot (SLOT5_PCIX) | 20 chassis intrusion header (INTRUDER) |
| 21 PCI Express 2.0 x16 card slot (SLOT4_PCIE) | 22 PCI card slot (SLOT3_PCI) |
| 23 PCI Express 2.0 x16 card slot (SLOT2_PCIE) | 24 front panel audio connector (FP_AUDIO) |
| 25 PCI Express x8 card slot, wired as x4 (SLOT1_PCIE) | 26 hard drive fan connector (FAN_HDD) |
| 27 memory module connectors (DIMM_1-8) | 28 memory fan connector (FAN_MEM) |

Figure 1-5 T5400-based Workstation System Board

Diagnostic Indicators

There are four lights on the front of the workstation. How these lights display indicate various states of the workstation. The lights can be off or green. When the workstation starts normally, the lights flash. If the workstation malfunctions, the pattern of the lights and also that of the power button help to identify the problem. These lights also indicate sleep states. Refer to Dell's Quick Reference Guide shipped with the workstation or to Dell's Web site.

1.3 Workstation Configurations

Workstation furniture options

The workstation can be positioned vertically or horizontally and used as a desktop unit or mounted in the following Honeywell furniture.

- Classic Console
- Z and EZ Console
- Icon Series Console
- Rittal Cabinet

When mounted in a cabinet, the enclosure uses the following amount of space:

- Slide Mount: 5U (8.75") of rack space
- Fixed Mount: 4U (7.00") of rack space



Figure 1-6 T5400-based Workstation for Cabinets and Consoles

Electronics module

The peripheral electronics assemblies in the workstation are based upon either the Peripheral Component Interconnect (PCI) bus, PCI Express (PCI-E), PCI Extended (PCI-X), or USB 1.1/2.0 protocols.

Storage and media devices

The standard mass storage for the workstation is one or two 80 GB or larger SATA hard drive. Except for the floppy drive, all mass storage and removable media devices connect using SATA depending on the configuration. The system has one SATA connection for the DVD±RW drive and two serial ATA (SATA) ports for mass storage. For the Icon Series Console, external media devices connect to the USB 2.0 ports.

Features

The workstation comes with the following features:

Workstation with a Non-Raid Platform – Tab 100

- CPU: 2.33GHz Xeon Stepped E5410 processor (single quad-core), 1333MHz, 2X6MB or greater L2 Cache
- Front Ports: 2-USB 2.0, 1-Stereo in/out (headphone), 1-Microphone, 1-IEEE connector
- Rear Ports: 2-serial (9-pin) , 1-parallel (DB25), 1-RJ45, 2-PS/2, 2-Stereo in/out (headphone), 6 USB 2.0 (5 back and 1 internal)
- Slots: 2-PCI-E x 16 graphics slot, 1-PCI-E x 8 wired as x 4, 1-32-bit/33MHz PCI, and 2-64-bit/100MHz PCI-X
- Audio: Integrated ESB2 AC'97/High Definition Digital Controller
- LAN: Integrated Broadcom Gigabit Ethernet Controller
- Power: 875 watts power factor correcting power supply
- Memory: 1GB (2x512MB), 667MHz, FB DDR2 ECC SDRAM, expandable to 2GB, 3GB, or 4GB
- Media Drive: SATA CDRW/DVDRW(±) dual layer driver w/Sonic Record Now
- Hard drive: Serial ATA, 1-80GB or larger, 10K RPM
- Removable drive: 3.5" 1.44MB floppy
- Video: 256MB NVidia QuadroFX570 dual video controller card
- Software: Microsoft Windows XP 32-bit Professional Service Pack 3 Operating System loaded as a downgrade from Windows Vista Ultimate

1. System Planning

1.3. Workstation Configurations

Workstation with a RAID1 Platform – Tab 200

- CPU: 2.83GHz Xeon Stepped E5440 processor motherboard (single quad-core), 1333MHz, 2X6MB or greater L2 Cache
- Front Ports: 2-USB 2.0, 1-Stereo in/out (headphone), 1-Microphone, 1-IEEE connector
- Rear Ports: 2-serial (9-pin) , 1-parallel (DB25), 1-RJ45, 2-PS/2, 2-Stereo in/out (headphone), 6 USB 2.0 (5 back and 1 internal)
- Slots: 2-PCI-E x 16 graphics slot, 1-PCI-E x 8 wired as x 4, 1-32-bit/33MHz PCI, and 2-64-bit/100MHz PCI-X
- Audio: Integrated ESB2 AC'97/High Definition Digital Controller
- LAN: Integrated Broadcom Gigabit Ethernet Controller
- Power: 875 watts power factor correcting power supply
- Memory: 1GB (2x512MB), 667MHz, FB DDR2 ECC SDRAM, expandable to 2GB, 3GB, or 4GB
- Media Drive: SATA CDRW/DVDRW(±) dual layer driver w/Sonic Record Now
- Hard drive: Serial ATA, 2-80GB or larger, 10K RPM
- Removable drive: 3.5" 1.44MB floppy
- Video: 256MB NVidia QuadroFX570 dual video controller card
- Software: Microsoft Windows XP 32-bit Professional Service Pack 3 Operating System loaded as a downgrade from Windows Vista Ultimate

Additional options

- LCNP4M
- Single NIC
- Dual NIC
- 1GB Memory Modules (2x512MB)

Specific slot configurations

The dual video display is the default option for the workstation platform. The Quad video display option includes three screen and quad screen video capabilities. The following tables define the slot requirements for workstation configurations for TPS, FTE, and Experion systems. Those devices identified as *optional* are only present if you specifically ordered them.

Table 1-3 Slot Configuration for TPS Systems

Slot Identification	Device
<i>Single/Dual screen configuration</i>	
Slot 6 – PCI-X	
Slot 5 – PCI-X	LCNP4M
Slot 4 – PCI-E x 16	
Slot 3 – PCI	Empty or Power Adapter Harness
Slot 2 – PCI-E x 16	Nvidia Quardro FX570
Slot 1 – PCI-E x 4	
<i>Quad screen configuration (single card option)</i>	
Slot 6 – PCI-X	Single Intel PRO 1000 Ethernet
Slot 5 – PCI-X	LCNP4M
Slot 4 – PCI-E x 16	
Slot 3 – PCI	Empty or Power Adapter Harness
Slot 2 – PCI-E x 16	Matrox QID
Slot 1 – PCI-E x 4	

Table 1-4 Slot Configuration for Experion Systems

Slot Identification	Device
<i>Quad video configuration</i>	
Slot 6 – PCI-X	
Slot 5 – PCI-X	LCNP4M
Slot 4 – PCI-E x 16	
Slot 3 – PCI	Empty or Power Adapter Harness
Slot 2 – PCI-E x 16	Matrox QID
Slot 1 – PCI-E x 4	
<i>Single/Dual video configuration</i>	
Slot 6 – PCI-X	
Slot 5 – PCI-X	LCNP4M
Slot 4 – PCI-E x 16	
Slot 3 – PCI	Empty or Power Adapter Harness
Slot 2 – PCI-E x 16	Nvidia Quadro FX570
Slot 1 – PCI-E x 4	

Table 1-5 Slot Configuration for FTE Systems

Slot Identification	Device
Single/Dual screen configuration	
Slot 6 – PCI-X	Dual Intel Pro 1000 Ethernet
Slot 5 – PCI-X	LCNP4M
Slot 4 – PCI-E x 16	
Slot 3 – PCI	Empty or Power Adapter Harness
Slot 2 – PCI-E x 16	Nvidia Quadro FX570
Slot 1 – PCI-E x 4	
Quad screen configuration	
Slot 6 – PCI-X	Dual Intel Pro 1000 Ethernet
Slot 5 – PCI-X	LCNP4M
Slot 4 – PCI-E x 16	
Slot 3 – PCI	Empty or Power Adapter Harness
Slot 2 – PCI-E x 16	Matrox QID
Slot 1 – PCI-E x 4	

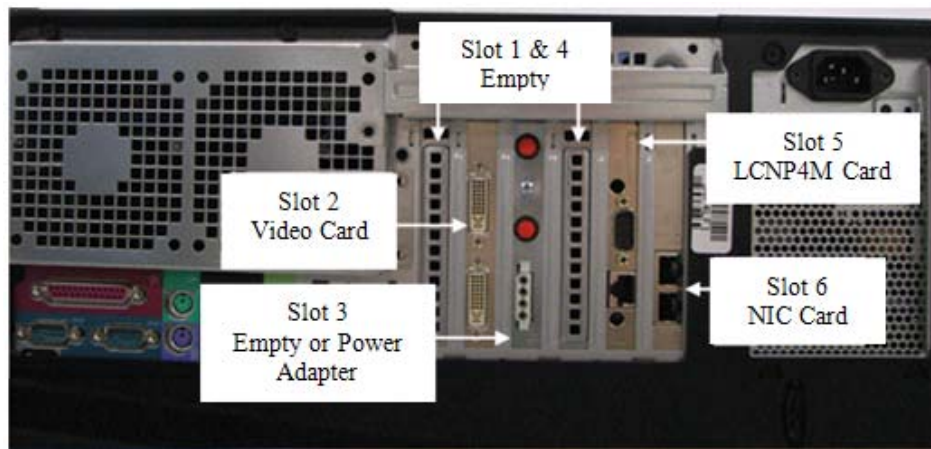


Figure 1-7 Example Slot Configuration

Memory configurations

The standard installed memory for the workstation is 1 GB. It can be expanded to 2, 3, or 4 GB by installing additional memory modules in the appropriate sockets as shown in the Figure 1-8. The memory devices must be from the same memory supplier and installed in matched pairs.

DIMM Socket	1 GB Memory (standard)	2 GB Memory	3 GB Memory	4 GB Memory
1	512 MB	512 MB	512 MB	512 MB
5	Empty	Empty	512 MB	512 MB
2	512 MB	512 MB	512 MB	512 MB
6	Empty	Empty	512 MB	512 MB
3	Empty	512 MB	512 MB	512 MB
7	Empty	Empty	Empty	512 MB
4	Empty	512 MB	512 MB	512 MB
8	Empty	Empty	Empty	512 MB

Figure 1-8 Memory Configuration

1.4 Finding Information for Your Workstation

Honeywell documentation

The following table lists Honeywell publications that may be useful when installing or operating your system.

Table 1-6 Honeywell Publications

Publication	Contains information on
ADP01: <i>Honeywell Peripheral Adapters</i>	Contains information for using the OEP/IKB adapter with workstation platforms that do not have the ISA card.
RE02: <i>Long-Haul Remote Systems Installation and Upgrade</i>	Contains information for using Honeywell's long-haul remote (300 feet) option.
FE05: <i>Fault Tolerant Ethernet Installation and Service Guide</i>	Contains information for installing and using FTE on a TPS or Experion PKS node.

Dell documentation

The following table lists Dell publications and other sources of information that will be useful when installing, operating, or servicing the workstation.

Table 1-7 Dell Publications

Publication	Contains information on	Available
Readme files	Last-minute updates about technical changes to your workstation or advanced technical-reference material for experienced users or technicians.	Drivers and utilities CD
System Information Label	System board connectors Location of system board components	Located on the inside cover of your workstation.
<i>Dell System Information Guide</i>	Warranty information Safety information	Packaged with the workstation www.dell.com

1. System Planning

1.5. Specifications

Publication	Contains information on	Available
<i>Setup and Quick Reference Guide</i>	How to set up my workstation How to care for my workstation Troubleshooting information How to run the Dell Diagnostics How to open my workstation cover How to locate other documentation	Packaged with the workstation www.dell.com
<i>Dell Precision™ Workstation T5400 User's Guide</i>	How to remove and replace parts Technical specifications How to configure system settings How to troubleshoot and solve problems	Product Documentation CD www.dell.com
<i>The Dell Precision™ T5400 Service Manual.</i>	Details on workstation configuration How to remove and replace parts	Product Documentation CD www.dell.com

1.5 Specifications

Environmental specifications for cabinets and consoles

The following table lists allowable operating environmental limitations for cabinets and consoles.

Description	Cabinets	Consoles
Ambient room temperature	5 – 30 deg C	5 – 30 deg C
Humidity	20 – 80% RH, non-condensing	20 – 80% RH, non-condensing
Operating vibration	0.012" P-P displ to 12.7 Hz, then 0.1g to 150 Hz	0.012" P-P displ to 20 Hz, then 0.15g to 150 Hz
Site induced shock	6g, 10 msec half-sine	10g, 10 msec half-sine

System specifications

Table 1-8 Specifications

Description	Specification
<i>Operating power requirements</i>	
AC Voltage	120 (90-135) Vrms 240 (180-265) Vrms
AC RMS Current	2.3 Arms 1.24 Arms
AC Power	279 Watts Max 279 Watts Max
<i>Weight and dimensions</i>	
Height	167.64 mm
Width	444.5 mm
Depth	452.12 mm
Weight	18.6 Kg depending on options
<i>Hard disk drive specifications (80GB)</i>	
DC 5 volt Power (typ)	5 Watts +/- 5%
DC 12 volt Power (typ)	12 Watts +10%, -8%
Other DC POWER	22 Watts Max, 10.0 Watts idle
AC Power	N/A
Height	25.4 mm
Width	101.6 mm
Depth	146.0 mm
Weight	0.6 kg

1. System Planning

1.5. Specifications

Removable media specifications

The following table lists specifications for removable media that can be used with the workstation platform.

Table 1-9 Specifications for Removable Media

Removable Media Power Requirements		
Description	Device Requirements	
	DVD+RW * EIDE/USB2.0	Zip Drive * SCSI/USB 2.0
DC 5 volt Power	6.0 Watts	5.0 Watts
DC 12 volt Power	21.6 Watts	N/A
Other DC POWER	N/A	N/A
AC Power	N/A	N/A
AC RMS Current	N/A	N/A
AC Power	N/A	N/A
Height	40.64 mm	41.5 mm
Width	146.05 mm	148 mm
Depth	14.22 mm	163 mm
Weight	1.20 kg	.067 kg
* Media drives mounted on Icon Series with an IDE to USB or SCSI to USB adaptor.		

Additional specifications

Refer to the workstation user guide for additional technical specifications and the vendor documentation for specifications on the peripheral devices.

Regulatory and safety compliance



WARNING

Honeywell does not claim Safety Compliance or Electromagnetic Compatibility (EMC) Compliance for system equipment configurations that have not been described in this manual as standard system configurations. Any equipment configuration other than that described in this publication decertifies the Safety and EMC compliance of this product.

2. Platform Installation

2.1 Introduction

Overview

This section contains procedures for installing and cabling the workstation in Honeywell consoles and cabinets.

Installation tasks

The specific tasks you need to perform will vary depending upon the type of furniture in which you are installing the workstation. The following table lists the major platform installation tasks.

Table 2-1 Major Platform Installation Tasks

Task		For more information see . . .
✓	Be aware of all power and grounding requirements for your furniture.	Specific site requirements Section 2.2, Power and Grounding Requirements <i>TPN System Installation (SW20-600)</i> , Section 4, "System Grounding"
✓	Verify pinning for external media drives is correct	Section 2.3, Pinning Media Drives
✓	Install the workstation in the furniture	Section 2.4, Install Workstation into the Icon Series Console Section 2.5, Install Workstation in Z and EZ Console Section 2.6, Install Workstation in Classic Console Section 2.7, Install the Workstation in Cabinets
✓	Connect device cables	Section 2.9, Connecting Cables
✓	If you are remoting the workstation, connect the remote components.	For short haul remote: <i>Remote Peripheral Systems Installation and Upgrade (RE01)</i> . For long haul remote: <i>Long-Haul Remote Systems Installation and Upgrade (RE02)</i> .

2.2 Power and Grounding Requirements

Grounding for workstation based nodes

The ground connection is made through the third wire in the AC power cord.

Grounding consoles and cabinets

The *TPN System Installation* manual, Section 4, “System Grounding” contains information on grounding furniture, including the following:

- Ground Wiring Overview
- Grounding LCN Cabinets and Stations
- Cabinet Logic Ground
- Grounding LCN Cables
- Grounding Stations in Z-Console Furniture

AC power warning



WARNING

The power supply circuit is connected to AC power when the power cable is connected. The power control switch on the front panel only enables the power supply circuit outputs.



ATTENTION

It is strongly recommended that the power cord be connected to a clean power source with backup such as an Uninterruptible Power Source (UPS).

Selecting the correct power setting

The workstation platform automatically senses the power supply therefore there is no positioning switch to be set.

2.3 Pinning Media Drives

Configure SCSI address for removable media drives

The workstation does not use a SCSI card option. If installing in an Icon Series Console, a SCSI to USB adapter may be used for a SCSI interface.

2.4 Install Workstation into the Icon Series Console

Use the following procedure to install the system workstation, air dam, and air gasket into the electronics enclosure of the Icon Series console.

Note: In the following procedure, “left” and “right” refer to viewing the electronics enclosure from the rear.

Step	Action
1	Place the workstation in the electronic enclosure with the back (connectors) of the workstation facing the left side of the enclosure. Make sure the workstation is centered and pushed up against the two elastomer bumpers at the bottom of the enclosure.
2	Fasten the tie-down straps into the slots of the enclosure cradle. Notice that there are multiple slots for the ends of the tie-down straps. The straps may be moved inward or outward depending on the size of the workstation. Attach the straps on the inward slots. Next, adjust the strap lengths so when the fastener is latched the straps are snug against the workstation.
3	Remove the adhesive from the air gasket (“L” shaped foam) and place it on top of the workstation as shown in Figure 2-2 on page 44. The adhesive keeps the gasket in place. The gasket stops exhaust air from re-circulating around the workstation.
4	Remove the adhesive from the air dam (clear flat sheet with two chamfered corners). Cover the ventilation slots at the bottom of the electronic enclosure near the front side of the workstation, with the chamfers downward. For the T5400 this is the right side. The dam causes cooling air to be drawn down from the back panel area (refer to Figure 2-1).
5	Test that the electronics enclosure can be closed and opened without binding. With some workstations, it may be necessary to trim or augment the air gasket.

2. Platform Installation

2.4. Install Workstation into the Icon Series Console

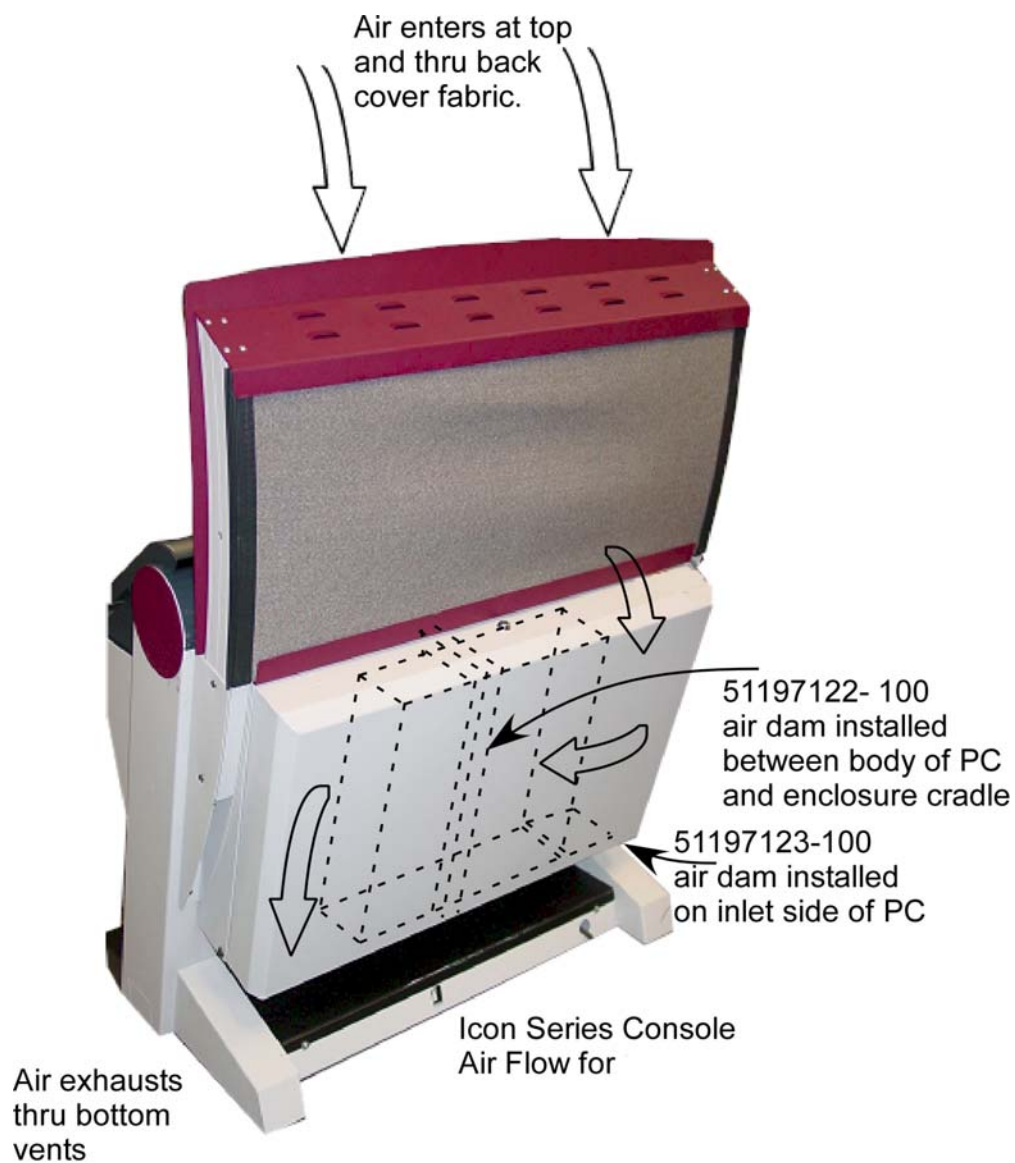


Figure 2-1 Air flow and Air Dam Arrangement for T5400-based Workstation

Connect system cables

Use the following procedure to connect the cables to the workstation.

Step	Action
1	If USB mouse and keyboard are not being used, skip this step. Connect the USB mouse and USB keyboard extension cables to the USB connectors.
2	Connect the audio cable from the amplifier to the audio (green) port.
3	If you are not using Fault Tolerant Ethernet (FTE), connect the Ethernet cable to the RJ-45 connector on the Network Interface Card on the back of the PC by the USB ports (Motherboard Ethernet connection). If you are using FTE, you will need to connect the FTE cables according to the instructions in the <i>FTE Installation and Service Guide</i>.
4	For Dual Video adapters, follow steps 5 and 6. For Quad Video adapters, follow steps 7 and 8.
5	For certain dual video adapters, a DVI-to-VGA converter will be required. The converter could be pre-installed. The second video channel is already VGA so a converter is not required. Note: For certain video adapters, DV1-D connection to the monitors can be used, if needed.
6	Connect the video cables to the analog video connectors on the video adapter. If you are in doubt about which output is #1 or #2, simply try one combination; this can be reversed later. Proceed to step 9.

2. Platform Installation

2.4. Install Workstation into the Icon Series Console

Step	Action
7	For certain quad adapters, connect the digital to analog converters to each of the adapter cables, as shown. Note: For certain video adapters, DVI-D connections to the monitors can be used, if needed. 

2. Platform Installation

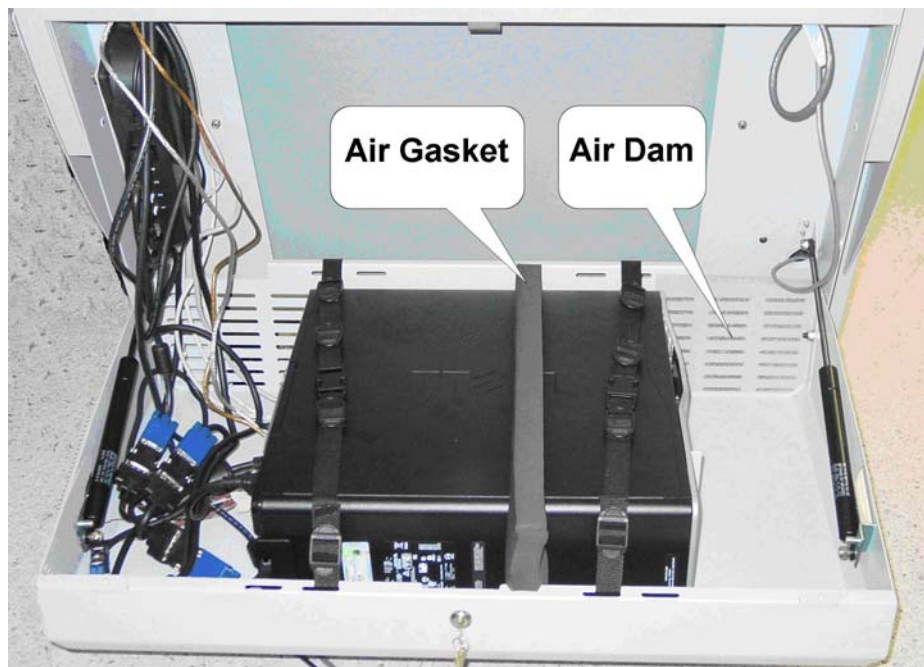
2.4. Install Workstation into the Icon Series Console

Step	Action
8	Connect the video adapter cable to the connector on the Quad video adapter/board. (This photograph shows this connection with the workstation outside of the enclosure on a desk.) 
9	Check the other ends of all the video cables to make sure they are connected to the appropriate connector on the monitors.
10	If your console has an OEP, connect the Serial cable from the OEP adapter to the COM1 port of the workstation.
11	The LCN MAU, if present, is pre-installed in the treadle area. Connect the LCN MAU cable to the MAU connector on the LCNP4 card if one is present. If other networks are being used, configure their MAU and cables appropriately.
12	Connect the one or two USB Hub cables to the workstation USB port(s).
13	Connect the workstation power cable to the workstation.
14	Arrange all cables so the enclosure can be raised and lowered without pinching, jamming, or straining any cable.

2. Platform Installation

2.4. Install Workstation into the Icon Series Console

Step	Action
15	The lower valance of the back panel has ten 2 inch holes. These are for both cable passage and air flow. Four hole plugs are also present. Install or move the hole plugs to the left side of the lower valance, into the left side holes. The left side holes are not filled with cables.  INSTALL HOLE PLUGS IN THESE HOLES IF THEY ARE NOT FILLED WITH CABLING KEEP THESE HOLES FREE FOR AIR INTAKE
16	Close the enclosure, securing it with the lock and, if necessary, shipping screws.



**Figure 2-2 Icon Series Console Electronic Enclosure
(loaded with workstation)**

2.5 Install Workstation in Z and EZ Console

Overview

This section provides instructions for installing a newly purchased T5400-based Honeywell workstation in a newly purchased Honeywell Z or EZ-console. The workstation has arrived from the Honeywell factory packaged in the Dell OEM shipping carton. This procedure assumes that a new Z or EZ console was shipped from the Honeywell factory with workstation hardware pre-installed.

Z and EZ mounting requirements

The workstation is installed in a fixed 15-degree reclining position in the upper or lower furniture bay of Z and EZ console furniture.

Install workstation in Z or EZ console

Use this procedure to install the workstation in the Z or EZ console furniture bay.

Step	Action
1	From the rear of the console, loosen the two fasteners on the rear cover. Remove the rear cover and set it aside. 
2	Slide the furniture bay cover off the mounting railings.

2. Platform Installation

2.5. Install Workstation in Z and EZ Console

Step	Action
3	Drape the installed cable harnesses over the sides of the module mounting brackets such that the console mounting rails are clear of any installed cables. 
4	Remove the rubber feet from the bottom and side of the workstation. 

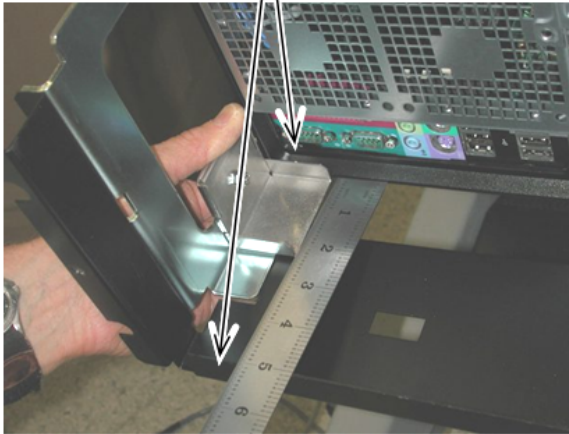
2. Platform Installation

2.5. Install Workstation in Z and EZ Console

Step	Action
5	Place the front of the workstation on the mounting railings, and slide the workstation into the furniture bay until the distance between the rear edge of the workstation and the vertical return flange on the rear cover retaining bracket is approximately 127mm [5"].  A photograph showing a person's hands sliding a black workstation into a furniture bay. The workstation is being positioned on mounting railings. The person is wearing a white shirt and a watch. The furniture bay is white and has a black front panel. The workstation is black and has a monitor and keyboard. The person is sliding the workstation into the bay until it is flush with the front panel.

2. Platform Installation

2.5. Install Workstation in Z and EZ Console

Step	Action
6	Install the left retaining bracket over the lower edge of the side mounting rail as shown below. Make sure the distance between the back panel of the workstation and the lip of the rear cover retaining bracket is five (5) inches as shown below.  Secure the bracket by tightening the M5 hex locknut.  In a similar manner, position and attach the right retaining bracket.

2. Platform Installation

2.5. Install Workstation in Z and EZ Console

Step	Action
7	Position the cable harnesses such that they are routed across the top of the workstation and insure that the cables lay flat and are not crossed. Slide the enclosure over the workstation and rail assembly.  
8	Go to Section 2.9, Connecting Cables, after connecting all cables return here to replace the rear cover.
9	Replace the rear cover and tighten both fasteners. Installation procedure is complete. Go to Section 3.1 to start up the workstation.

2. Platform Installation

2.6. Install Workstation in Classic Console

2.6 Install Workstation in Classic Console

Overview

This section provides instructions for installing a newly purchased T5400-based Honeywell workstation in a newly purchased Honeywell Classic Console. The workstation has arrived from the Honeywell factory packaged in the Dell OEM shipping carton. This procedure assumes that a new classic console was shipped from the Honeywell factory with workstation hardware pre-installed.

Classic console mounting requirements

The workstation is installed in the lower mounting location in a horizontal position in the Classic Console using a slide mount or fixed mount option. The following table describes the requirements for these mounting options.

Table 2-2 Slide and Fixed Mounting Requirements

Option	Rack Space Needed	Maximum T5400s	Mounting Requirements
Slide	5U (8.75 inches)	1 per console	Must be mounted on bottom of console. The 3U of unused space must have a 3U blank front panel and air baffle installed.
Fixed	4U (7.00 inches)	2 per console	If only one workstation is installed, it must be mounted in the console's lower mounting location with a 4U blank front panel and air duct baffle installed in the unused space.

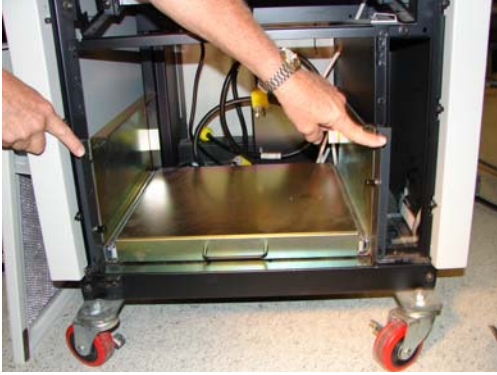
Install workstation in classic console using the fixed mount option

Use this procedure to install the workstation in Classic Console furniture using the fixed mount option.

Step	Action
1	Remove the console door to access the console enclosure.
2	Determine an appropriate location for the workstation by referring to the requirements listed in Table 2-2.

2. Platform Installation

2.6. Install Workstation in Classic Console

Step	Action
3	If the retaining brackets are attached, loosen the two captivated fasteners on each retaining bracket and remove them. This picture illustrates a slide mount configuration, but the brackets are similar for the fixed mount configuration. 
4	Verify that the console mounting rails are clear of any installed cables. 

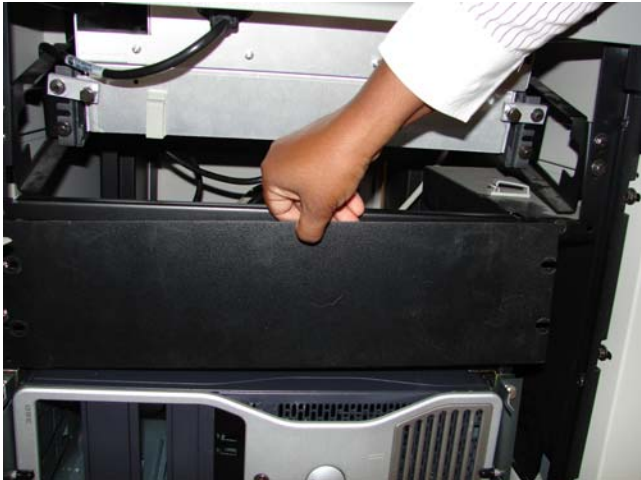
2. Platform Installation

2.6. Install Workstation in Classic Console

Step	Action
5	Slide the workstation all the way into the console enclosure between the mounting railings as shown. Replace the two retaining brackets and tighten the four captivated fasteners. 
6	If you are installing a second workstation in the console, repeat steps 2 through 5 for the second workstation, then go to Section 2.9, Connecting Cables. If you are not installing a second workstation and if an air duct baffle and blank front panel is not already installed continue to the next step.
7	Install a 4U high air duct baffle against the console right mounting rail. 

2. Platform Installation

2.6. Install Workstation in Classic Console

Step	Action
8	Attach the blank front panel to the console rails using two hex head screws and two washers, making sure the flat side of the washer touches the screw head and the washer teeth touch the panel.  Note: Installing a blank front panel ensures that the airflow within the console allows proper cooling of the workstation.
9	Go to Section 2.9, Connecting Cables.

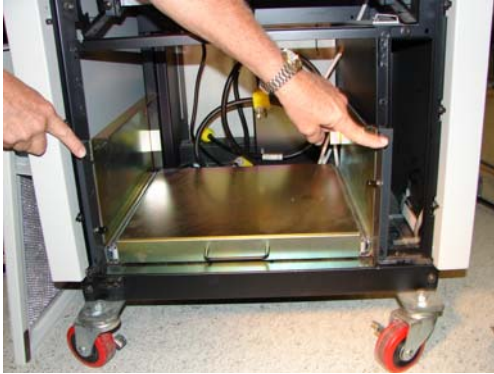
Install workstation in classic console using slide mount option

Use this procedure to install the workstation in classic console furniture using the slide mount option.

Step	Action
1	Remove the console door to access the console enclosure.

2. Platform Installation

2.6. Install Workstation in Classic Console

Step	Action
2	If the retaining brackets are attached, loosen the two captivated fasteners on each retaining bracket and remove them. 
3	Verify that the slide mount bottom and sides are clear of any installed cables.
4	Using the mounting tray handle, fully extend the tray from the console.
5	Place the workstation on the mounting tray and slide the tray back into the console.
6	Replace the two retaining brackets and tighten the four captivated fasteners.
7	If an air duct baffle and blank front panel are already installed, go to Section 2.9, Connecting Cables. If an air duct baffle and blank front panel are not installed continue to the next step.

2. Platform Installation

2.6. Install Workstation in Classic Console

Step	Action
8	Install a 3U high air duct baffle against the console right mounting rail. 
9	Attach the blank front panel to the console rails using two hex head screws and two washers, making sure the flat side of the washer touches the screw head and the washer teeth touch the panel.  Note: Installing a blank front panel ensures that the airflow within the console allows proper cooling of the workstation.
10	Go to Section 2.9, Connecting Cables.

2. Platform Installation

2.7. Install the Workstation in Cabinets

2.7 Install the Workstation in Cabinets

Overview

This section provides instructions for installing a newly purchased T5400-based Honeywell workstation in a newly purchased Honeywell cabinet (800mm deep equipment cabinet, Honeywell model number MP-C8LCB1 or 1000mm deep equipment cabinet, Honeywell model number MP-C1MCB1). The workstation has arrived from the Honeywell factory packaged in the Dell OEM shipping carton. This procedure assumes that a new Honeywell cabinet was shipped from the Honeywell factory with workstation hardware pre-installed.

The workstation mounts horizontally in a cabinet. The human interface is accessed locally in the cabinet or remotely using Honeywell's remote access options.

Cabinet mounting requirements

The various devices mounted in the cabinet require RETMA standard cabinet openings as specified in the following table.

Option	Rack Space Needed	Mounting Requirements
Slide mounted T5400	5U 8.75 inches	Any unused space must have a blank front panel and air duct baffle installed in the unused space to ensure proper airflow in the cabinet. Due to thermal constraints, a maximum of six (6) computing nodes can be installed in an 800mm deep cabinet and a maximum of nine (9) computing nodes can be installed in a 1000mm deep cabinet.
Fixed mounted T5400	4U 7.00 inches	
Human interface: 8 port KVM switch; 15-inch LCD; keyboard & cursor device	1U 1.75 inches	Mount in rack space 13U on telescopic slides

Unused cabinet spaces



ATTENTION

To ensure proper workstation cooling and airflow through the cabinet, all unused rack mount locations must have a blank front panel and air duct baffle installed.

All unused rack mount locations must have blank front panels and air duct baffles installed to ensure proper cooling. These are available in four height options. The following table shows the four height options and the corresponding part and tab numbers each.

Table 2-3 Air Duct Baffle and Blank Front Panel Height Options

Height Option	Part	Part Number	Tab number
1U	Blank front panel	51201248	-100
	Air duct baffle	51303521	-100
2U	Blank front panel	51201248	-200
	Air duct baffle	51303521	-200
3U	Blank front panel	51201248	-300
	Air duct baffle	51303521	-300
4U	Blank front panel	51201248	-400
	Air duct baffle	51303521	-400

Install workstation in cabinet using the fixed mount option

Use this procedure to install the workstation in the cabinet using the fixed mount option.

Step	Action
1	Open the cabinet door to access the inside of cabinet.

2. Platform Installation

2.7. Install the Workstation in Cabinets

Step	Action
2	If the retaining brackets are attached, loosen the two captivated fasteners on each bracket and remove them. 
3	Verify that the cabinet mounting rails are clear of any installed cables.
4	Slide the workstation all the way into the cabinet enclosure between the mounting railings until it touches the return flanges formed out of the mounting rails.
5	Replace the two retaining brackets and tighten the four captivated panel fasteners. Next, adjust the stops in the retaining brackets such that the vinyl caps contact the workstation's front bezel. Tighten the stops.
6	Go to the "Install air duct baffles and blank panel front covers in cabinet" procedure in this section.

Install workstation in cabinet using the slide mount option

Use this procedure to install the workstation in the cabinet using the slide mount option.

Step	Action
1	Open the cabinet door to access the inside of cabinet.
2	If the retaining brackets are attached, loosen the two captivated fasteners on each bracket and remove them.
3	Verify that the slide mount bottom and sides are clear of any installed cables.
4	Using the mounting tray handle, pull the tray to its fully extended and locked position. 

2. Platform Installation

2.7. Install the Workstation in Cabinets

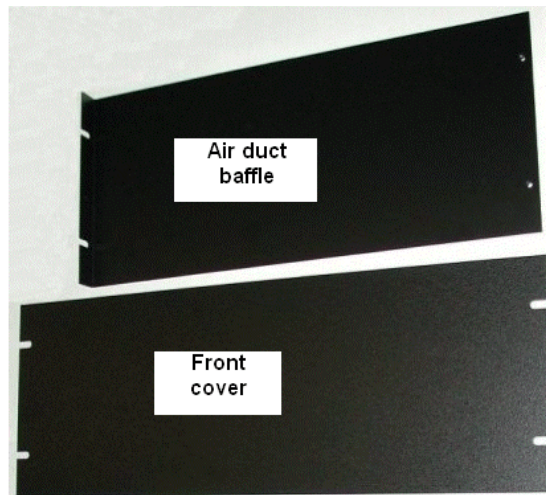
Step	Action
5	Place the workstation on the mounting tray so the bottom surface of the front bezel is aligned with the front surface of the slide tray and slide the tray back into the console. Note: A different workstation is shown in this photograph. 
6	Replace the two front retaining brackets and tighten the four captivated fasteners. Next, adjust the stops in the front retaining brackets such that the vinyl caps contact the workstation's front bezel. Tighten the stops.
7	From the rear of the cabinet, slide the workstation rear retainer bracket (L-shaped angle bracket) forward in its slot until it contacts the back surface of the workstation chassis. Tighten the lock nut.
8	Go to the "Install air duct baffles and blank panel front covers in cabinet" procedure in this section.

2. Platform Installation

2.7. Install the Workstation in Cabinets

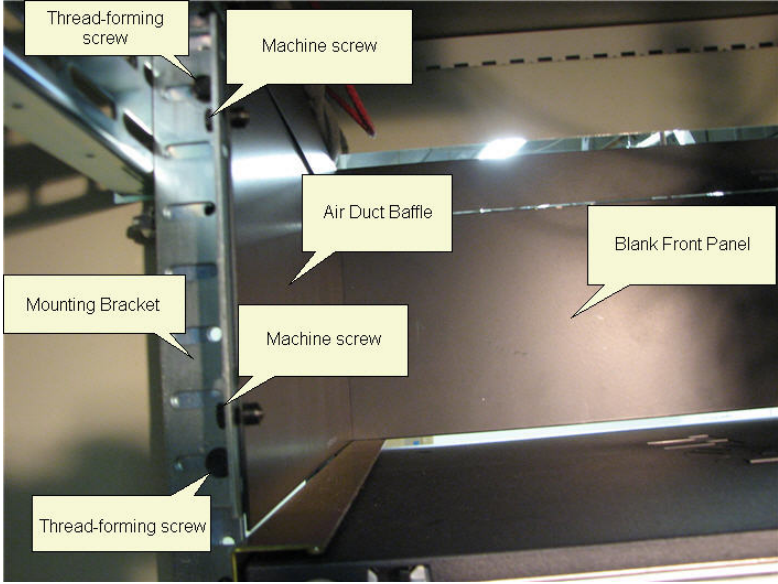
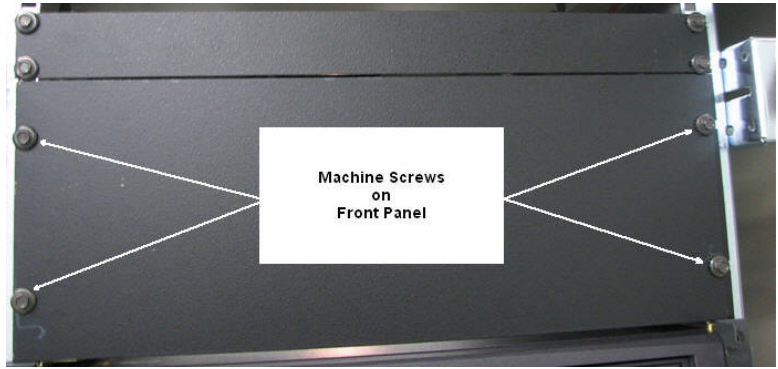
Install air duct baffles and blank panel front covers in cabinet

Use this procedure to install the air duct baffle and blank panel front cover in the cabinet. Installing these parts will insure that the airflow within the cabinet allows proper cooling of the workstations.



2. Platform Installation

2.7. Install the Workstation in Cabinets

Step	Action
1	Place the air duct baffle inside the cabinet with the bent tab resting along the front of the right cabinet rail. 
2	Place the blank front panel across the front of the cabinet rails and attach it to the air duct baffle and right cabinet rail using two machine screws and two external tooth washers threaded into the two clip nuts and tighten the screws. 
3	Attach the other side of the blank front panel to the left cabinet rail using two machine screws and two external tooth washers threaded into the two clip

Step	Action
	nuts and tighten the screws.
4	Attach the rear mounting bracket to the rear cabinet rail using two thread forming screws and two external tooth washers and tighten the screws.
5	Attach the air duct baffle to the rear mounting bracket using two machine screws and two external tooth washers threaded into the two self clinching nuts. Tighten the screws.
6	Go to Section 2.9, Connecting Cables.

2.8 Installing Optional Components

Honeywell provides a number of other options that may be installed in the cabinet with your workstation. Each of the following options is packaged with installation instructions.

Component	Description
OEP/IKB adapter	Used to connect an OEP or IKB that uses an ISA connection to the workstation's serial port. See <i>Honeywell Peripheral Adapters (ADP01)</i>
Long-haul remote components	Used for remotng the workstation up to 300 feet from the cabinet. See <i>Long-Haul Remote Systems Installation and Upgrade (RE02)</i> .

The following figure illustrates a suggested placement of optional adapters in a console.

Note: An earlier model workstation is shown.

2. Platform Installation

2.9. Connecting Cables

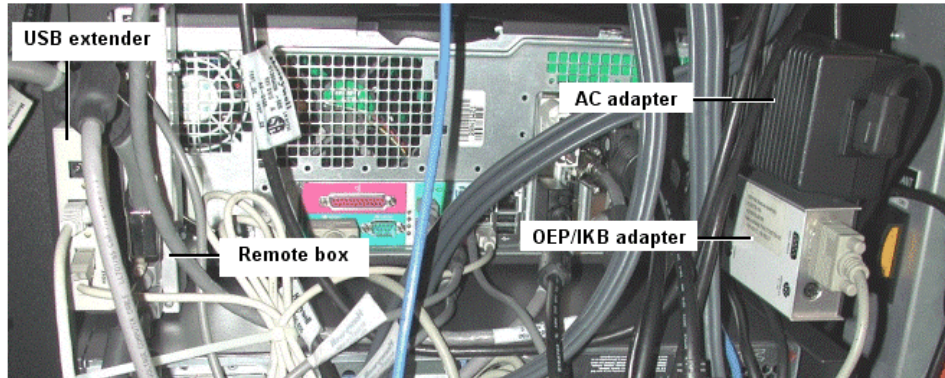


Figure 2-3 Suggested Placement of Optional Adapters

2.9 Connecting Cables

Connect cables to the workstation

This section contains information for connecting the power cord and cables to the workstation platform. Your configuration may not have all the card connections shown below.

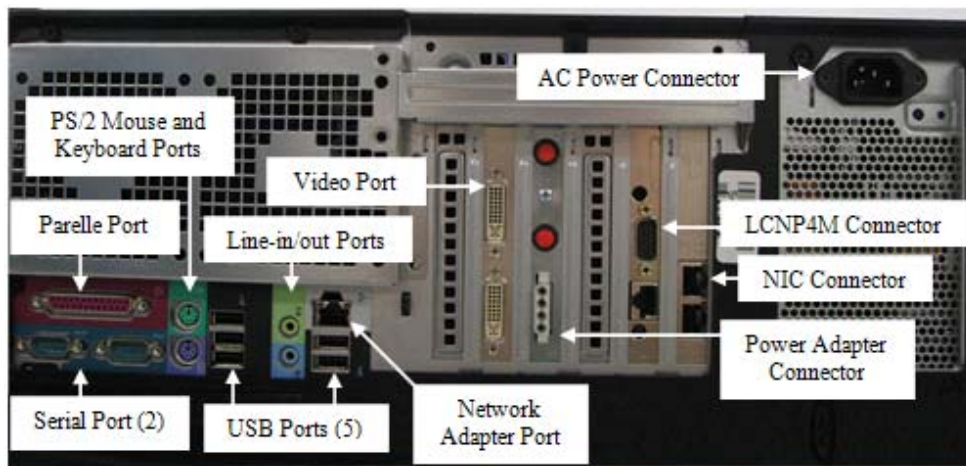


Figure 2-4 Connect Cables to T5400-based Workstation

Step	Action
1	Connect a parallel device, such as a printer, to the parallel port. If you have a USB printer, plug it into a USB port. Note: The integrated parallel port is automatically disabled if the workstation detects an installed card containing a parallel port configured to the same address. For more information, see System Setup Options.
2	Connect a serial device, such as a handheld device, to the serial port. If necessary, the address for this port can be modified through System Setup. See Table 2-5 for applicable serial port connections.
3	If you have a standard USB keyboard, plug it into a USB port. If you have a PS/2 keyboard, plug it into the PS/2 purple keyboard port.
4	If you have a standard USB mouse, plug it into a USB port. If you have a PS/2 mouse, plug it into the PS/2 green mouse port.
5	Use the back USB ports for devices that typically remain connected, such as a printer, mouse, keyboard, and touchscreen connections. It is recommended that you use the front USB ports for devices that you connect occasionally, such as flash memory keys, cameras, or for bootable USB devices.
6	Use the green line-out port to attach headphones and most speakers with integrated amplifiers. On workstations with a sound card, use the connector on the card.
7	Use the blue line-in port to attach a record/playback device such as a cassette player, CD player, or VCR. On workstations with a sound card, use the connector on the card.

2. Platform Installation

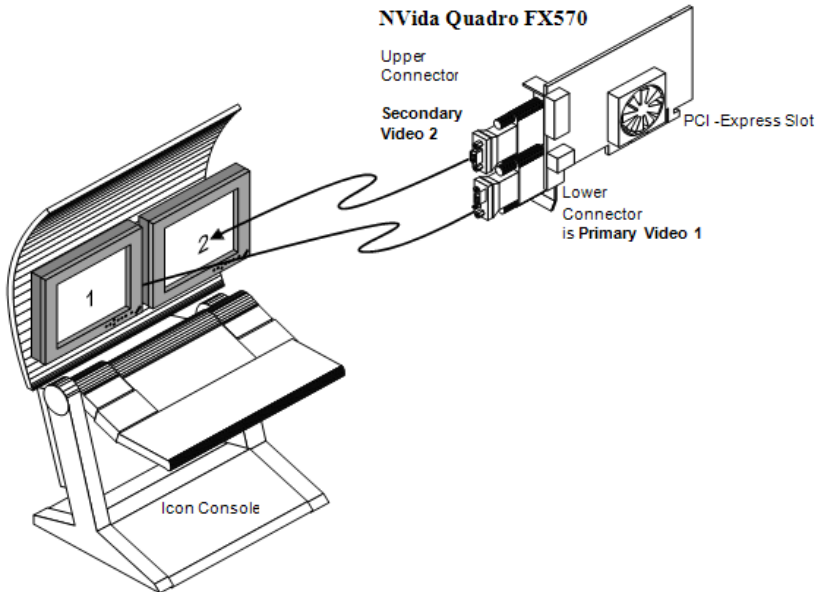
2.9. Connecting Cables

Step	Action
8	To attach your workstation to a network or broadband device, connect one end of a network cable to either a network jack or your network or broadband device. Connect the other end of the network cable to the network adapter port on your workstation. A click indicates that the network cable has been securely attached. Note: Do not plug a telephone cable into the network port. On workstations with an additional network connector card, use the connectors on the card and on the back of the workstation when setting up multiple network connections (such as a separate intra and extranet). It is recommended that you use Category 5 wiring and connectors for your network. If you must use Category 3 wiring, force the network speed to 10 Mbps to ensure reliable operation.
9	Connect the AC power cord.

Connect remaining cables and power

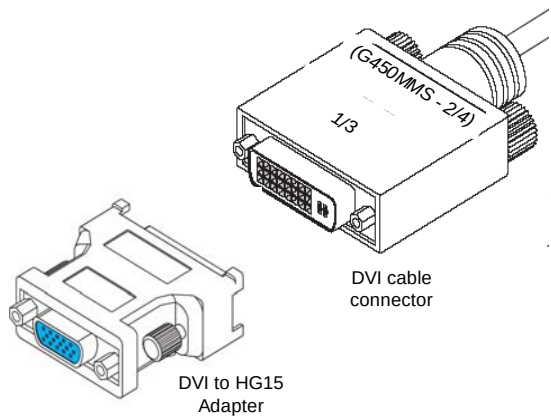
Use this procedure to connect the remaining cables to the workstation.

Step	Action
1	If you have an IKB that uses the OEP/IKB adapter: • Connect the PS/2 keyboard cable from the OEP/IKB adapter to the keyboard PS/2 port. • Connect the DB9 serial extension cable to the serial port.
2	If you are not using Fault Tolerant Ethernet (FTE), connect the Ethernet cable to the RJ-45 connector on the Network Interface Card. If you are using FTE, you will need to connect the FTE cables according to the instructions in the <i>FTE (FE05) Installation and Service Guide</i>.
3	Connect the LCN MAU cable to the MAU connector on the LCNP4M card.
4	Connect any USB devices or Hubs to the USB ports, including the USB IKB if you are using one.

Step	Action
Connecting monitors (single/dual video card)	
5	Connect the DVI to HD15 adapter to the lower digital connector and connect the primary video cable to that adapter. If you are connecting two video devices, connect a DVI to HD15 adapter to the upper digital connector and connect the secondary video cable to the top adapter. 

2. Platform Installation

2.9. Connecting Cables

Step	Action
Connecting monitors (quad video card (Icon Series only))	
6	Connect the DVI to HD15 adapters to the Quad video DVI Cable by adding the VGA adapters (HD15) to each video connector in the DVI cable as shown.  DVI to HG15 Adapter DVI cable connector
7	Connect the Quad video DVI Cable to the video connector on the video card.
8	Connect the video cables according to the information in the following "Connect Icon Series Monitors" procedure.
9	Secure any loose cables, and verify that all cables have proper strain relief.

Connect Icon Series Monitors

For systems that have three or four video outputs, the workstation uses the Quad video DVI Cable connected to a quad video card. You must connect the monitors to the correct channels identified in Table 2-4. See also Figure 2-5 and Figure 2-6.

Note a DVI to HG15 adapter must be installed on each of the Quad video DVI Cable prior to attaching the video cables from the displays.

Table 2-4 Video Channels

Channel	Four Monitor Connection	Three Monitor Connection
1	DVI Cable connector labeled 1/5	DVI Cable connector labeled 1/5
2	DVI Cable connector labeled 2/6	DVI Cable connector labeled 2/6
3	DVI Cable connector labeled 3/7	DVI Cable connector labeled 3/7
4	DVI Cable connector labeled 4/8	Not used

2. Platform Installation

2.9. Connecting Cables

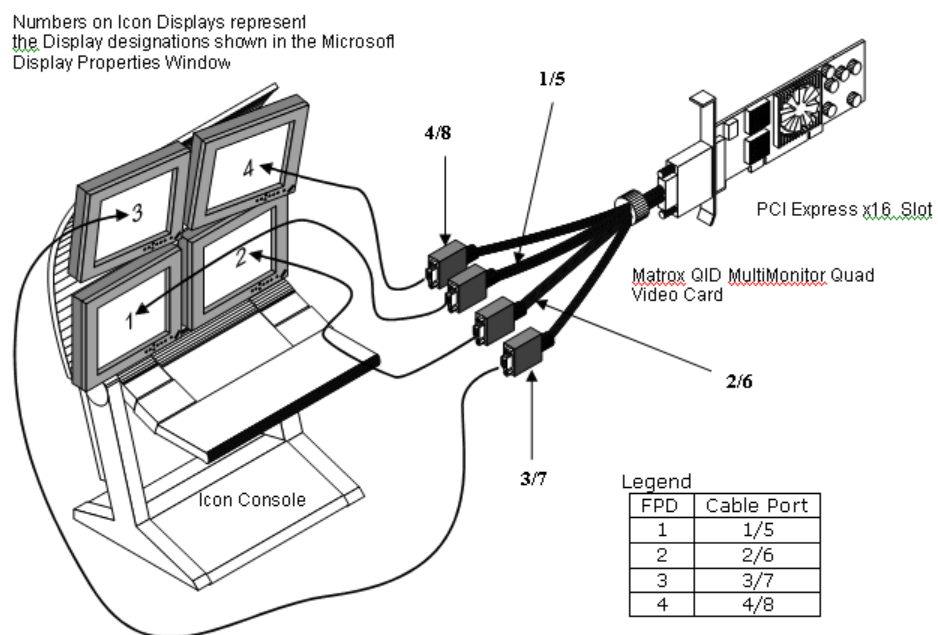


Figure 2-5 Four Monitor Configuration

2. Platform Installation

2.9. Connecting Cables

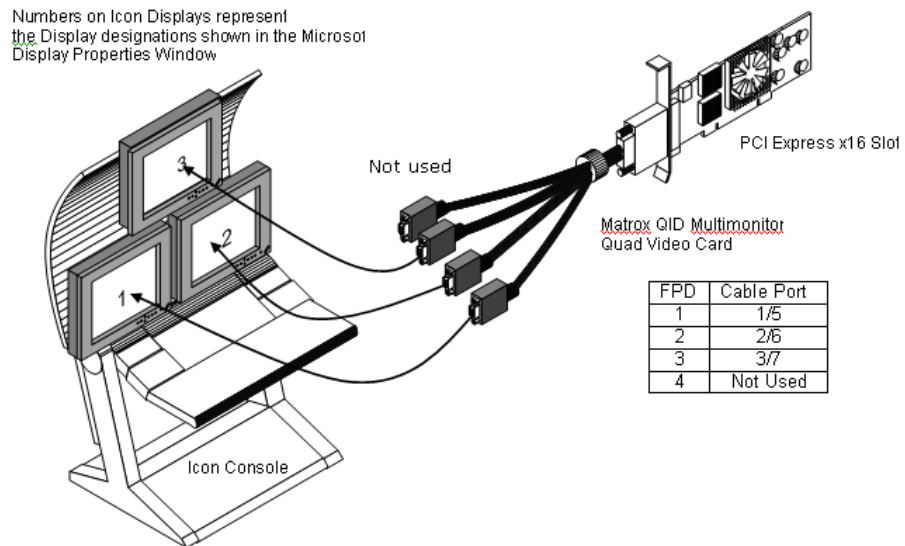


Figure 2-6 Three Monitor Configuration

2.10 Connecting Adapters

Workstation serial ports

The workstation has two standard 9-pin serial ports COM 1 and COM 2. See Figure 2-4 to view the serial ports. You must connect Honeywell devices to the correct serial port as defined in the following table.

Table 2-5 COM Port Connections

Honeywell Device	Connect to this Serial Port
OEP	COM1 (configured as COM3)
IKB	COM1 (configured as COM3)
Non-USB Touchscreen	COM2 (configured as COM4) Note: A USB port is used for a flat panel display touchscreen.

OEP/IKB Adapter Configurations

Purpose

Use the information and procedures in this section to connect the OEP/IKB adapter. The following block diagrams show the basic connections for the OEP/IKB adapter assembly used in different configurations.

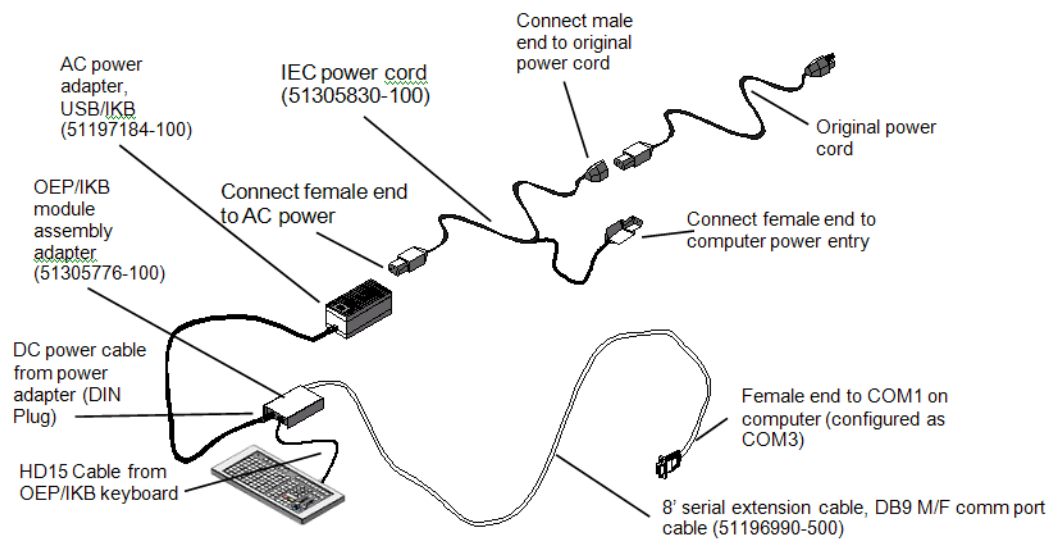


Figure 2-7 OEP/IKB Adapter Connections for OEP

2. Platform Installation

2.10. Connecting Adapters

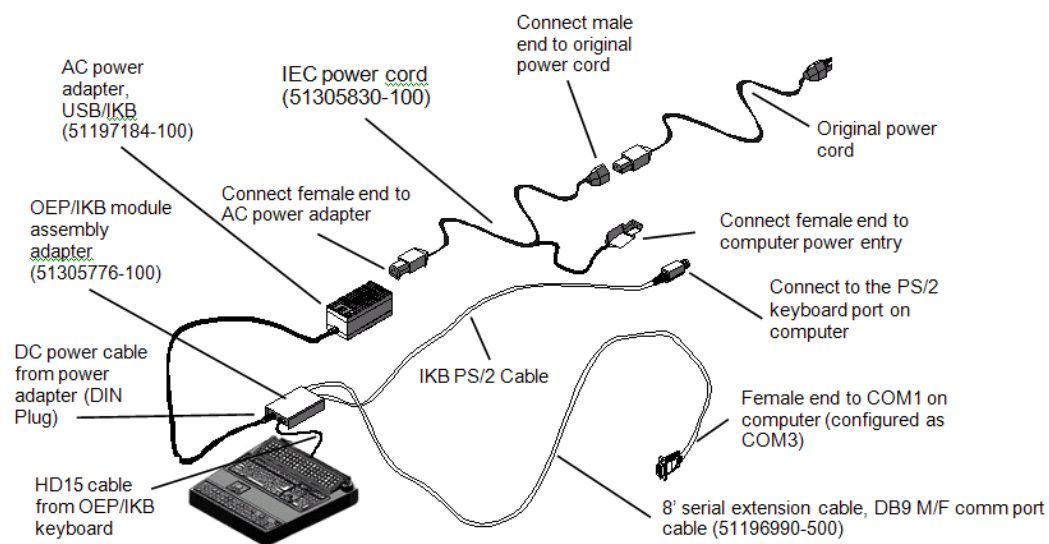


Figure 2-8 OEP/IKB Adapter Connections for IKB

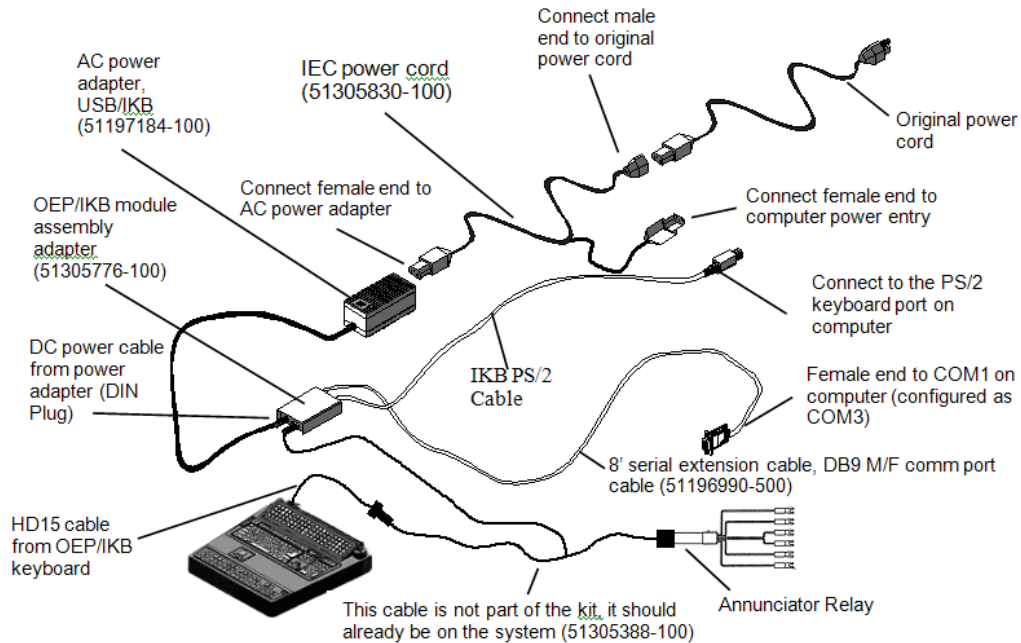


Figure 2-9 OEP/IKB Adapter Connections for IKB with Annunciator Relay

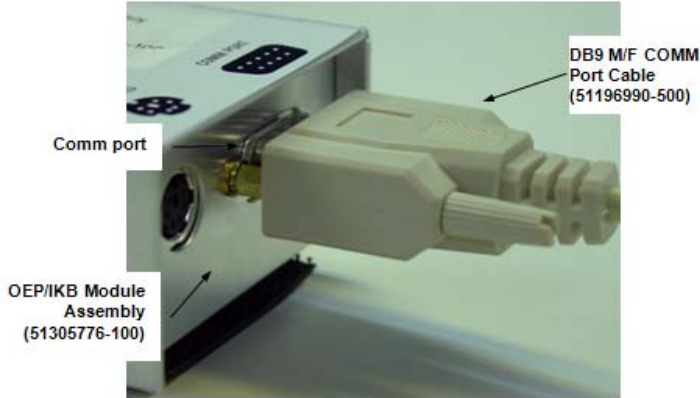
Install OEP/IKB adapter

Use this procedure to connect the OEP/IKB adapter.

Step	Action
1	Locate a safe location for the OEP/IKB adapter box and secure it using the supplied Velcro.
2	Locate a safe location to place the OEP/IKB power adapter box and secure it using the supplied Velcro.
3	If The IKB has a trackball connect it to the mouse PS/2 port on the workstation.

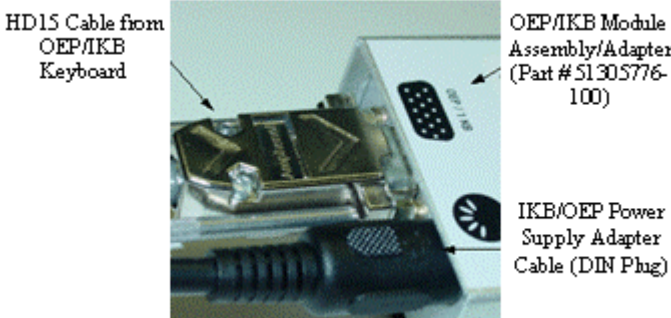
2. Platform Installation

2.10. Connecting Adapters

Step	Action
4	Locate and connect the male end of the DB9 M/F COM Port Cable (51196990-500) to the COM PORT on the OEP/IKB Module Assembly/Adapter (51305776-100). 
5	Secure the connection by tightening the thumbscrews located on the cable end.
6	Connect the female end of the DB9 M/F COM Port Cable (51196990-500) to COM 1 port located on the workstation's back panel and tighten the thumbscrews.
7	Insert the round DIN plug of the supplied IKB PS/2 cable (51305381-500) into the PS/2 Connector on the OEP/IKB Module Assembly (51305776-100). 

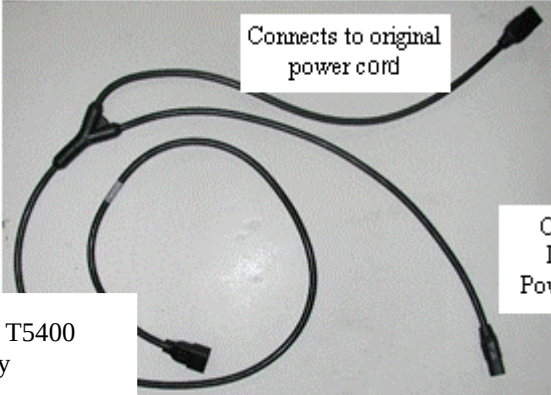
2. Platform Installation

2.10. Connecting Adapters

Step	Action
8	Connect the free end of the IKB PS/2 cable (51305381-500) to the Keyboard PS/2 Port on the workstation.
9	Take the DC power cable from the IKB/OEP Power Supply Adapter (51197184-100) and insert the round DIN plug into the Power Connector receptacle on the IKB/OEP Module Assembly/Adapter (51305776-100). 
10	Connect the existing OEP/IKB Keyboard HD15 Cable to the HD15 OEP/IKB receptacle located on the IKB/OEP Assembly/Adapter (51305776-100). Secure cable to the assembly/adapter by tightening the thumbscrews located on the cable's end. 

2. Platform Installation

2.10. Connecting Adapters

Step	Action
11	Connect the IEC power cord (51305830-100): • Connect one female end to the OEP/IKB power adapter (51197184-100). • Connect the other female end to the T5400 power entry. • Connect the male end to the original workstation power cord. 

Connect Touchscreen Power Adapter Cable

Purpose

Use the information and procedures in this section to connect the touchscreen power cable for the following touchscreen configurations:

- Touchscreen is connected to the touchscreen adapter box.
- Touchscreen is connected to the DB9 socket on the back of the monitor.
- Touchscreen is connected to the DB9 socket on the touchscreen frame.

Connect touchscreen adapter cable

Use this procedure to connect the supplied touchscreen adapter cable to the touchscreen device and the workstation.

Step	Action
1	Remove the existing DB9 touchscreen cable.

Step	Action
2	If your touchscreen uses the touchscreen adapter box, connect the end of the new touchscreen cable that does not have the 4-pin connector to the DB9 socket on the box as shown. 
3	If your touchscreen uses the DB9 socket on the back of the monitor, connect the end of the new touchscreen cable that does not have the 4-pin connector to the DB9 socket on the back of the monitor.
4	If your touchscreen uses the DB9 socket on the touchscreen frame, connect the end of the new touchscreen cable that does not have the 4-pin connector to the DB9 socket on the touchscreen frame. Notes: If 3 or 4 do not apply to your configuration you must upgrade to a flat panel display for touchscreen functionality. If you are mounting in a console the CRT cover needs to be removed.
5	Connect the other end of the touchscreen cable (end with the 4-pin connector) to the COM2 serial port on the workstation.
6	Connect the 4-pin connector to the remote power circuit assembly card on the workstation.

2.10. Connecting Adapters

3. Workstation Operation

3.1 Starting up your Workstation

Overview

The following table lists those tasks that must be performed before operating your workstation. This section contains step-by-step procedures for each of these tasks.

✓	Task
	Turn on power and check status
	Check LCNP status
	Set monitor resolution
	Monitors with the Matrox QID Card Use this procedure if you have a quad video configuration and are not using all four of the available video display connectors on the installed Matrox QID video card. Configure three monitors for Icon Series Consoles

Turn on power and check status

Step	Action
1	Press the power button on the front panel of the workstation.
2	Wait for the power light to become solid green.
3	If the power light does not become solid green, refer to the "Advanced Troubleshooting" section of the <i>Dell Precision™ Workstation T5400 User's Guide</i> .

3. Workstation Operation

3.1. Starting up your Workstation

Check LCNP status

If this is a TPS node with the LCNP4M board installed, verify that the LCNP passed self-test.

Step	Action
1	From the Start menu, select LCNP Status .
2	Verify that the LCNP status indicates Passed Self Test and the circle is green.
3	Verify that LCN Address appears in the LEDs field of the LCNP Status display.



REFERENCE

Refer to the LCNP Status section in the *LCNP Status User's Guide* for more information.

Set monitor resolution

When the workstation is initialized, the monitor configuration is established based on the following user input:

- CRT type monitors: 75 Hz
- FPD type monitors: 60 Hz

If, after initialization, you install a monitor type different than what was originally defined, you will need to adjust the monitor settings for optimal performance. See your specific monitor user guide for other recommended settings.

Step	Action
1	Select Start > Settings > Control Panel .
2	Double-click the Display Icon Series and select the Settings tab.
3	Under Screen area, drag the slider to the appropriate resolution.
4	Click Apply .

Monitors with the Matrox QID Card

Use this procedure if you have a quad video configuration and are not using all four of the available video display connectors on the installed Matrox QID video card.

Configure three monitors for Icon Series Consoles

Use this procedure if you have a quad video configuration and are not using all four of the available video display connectors on the installed video cards.



ATTENTION

If you are not using all available channels on the installed video cards, you must configure your monitors such that the monitors are connected to the lowest and highest channels.

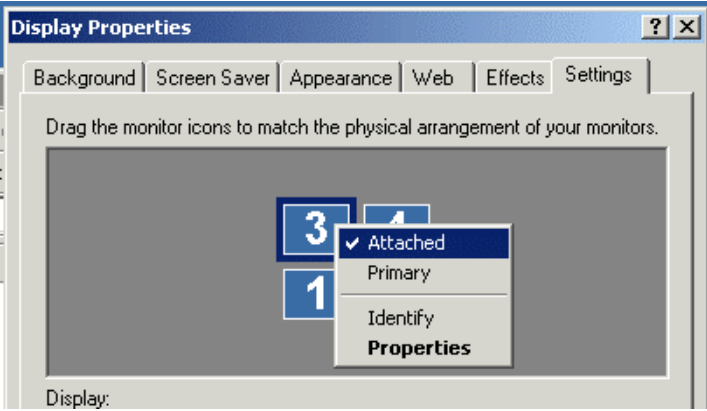
For example: If you are using 3 monitors with the two dual video cards installed, you must:

- Attach the monitors to channels 1, 2 and 4.
 - In the monitor settings, disable monitor 3 and leave monitors 1, 2, and 4 attached.
-

Step	Action
1	Verify that the monitors are physically attached to at least the highest and lowest video channels. See Figure 2-6, Three Monitor Configuration.
2	Select Start > Settings > Control Panel .
3	Double-click the Display Icon Series and select the Settings tab.
4	Right-click on monitor to be disabled (most likely number 3).

3. Workstation Operation

3.2. Network Connections

Step	Action
5	Select Attached to uncheck the option and disable the monitor. Click OK. 

3.2 Network Connections

Overview

Each workstation platform must be connected to an LCN network and/or an ETHERNET network.

Ethernet network

ETHERNET 10/100/1000 Base T connection is standard on the workstation platform. A Dual NIC option is available for FTE. The on board Ethernet connection must be disabled in the System BIOS prior to installation of the Dual Intel PRO 100/1000 controller. If you are using FTE, you must install the FTE cables according to the specific instructions in the *FTE Installation and Service Guide*.

LCN network

The connection to the LCN is made via an optional Local Control Network Processor LCNP4M (mid size) card. This card provides the communication path for the workstation to other LCN modules. The LCN Network Connection consists of an LCNP4M card, a MAU cable and the LCN MAU (Media Access Unit). The LCNP4, MAU and MAU cable are required to connect to the LCN Network. For the workstation the LCNP4M consists of a PCI/PCI-X Express 3.3 / 5.0 Volt LCNP4M medium size card, a MAU cable and the LCN MAU.

The LCN node address should be set to the address the customer requires and configured in the Network Configuration File (NCF). If the LCN address is not known then the node address should be set to zero (0). Setting the address to zero (0) allows the node to be connected to the LCN without the risk of an address conflict with some other node. This is consistent with the current LCN standard procedure.

Refer to *LCN System Installation*, Subsection 3.6 for the rules and techniques of installing an LCN cable system.

3. Workstation Operation

3.2. Network Connections

LCN connections

The LCN Cable A and Cable B connections are made through a single cable from the LCNP4M board to the LCN Media Access Unit (MAU) contained in a metal housing.

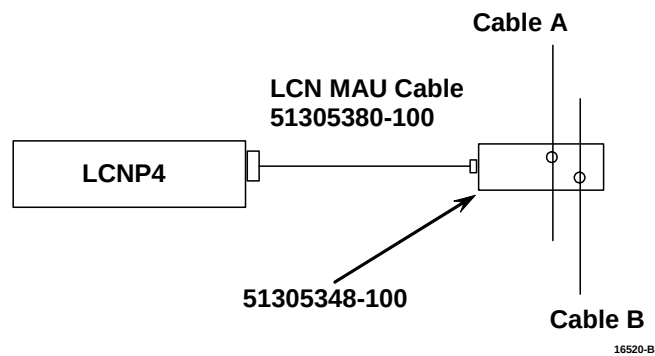


Figure 3-1 LCN MAU to LCN Cabling

MAU connection

Connect the MAU to both Cable A and Cable B coax T-connector as shown below.

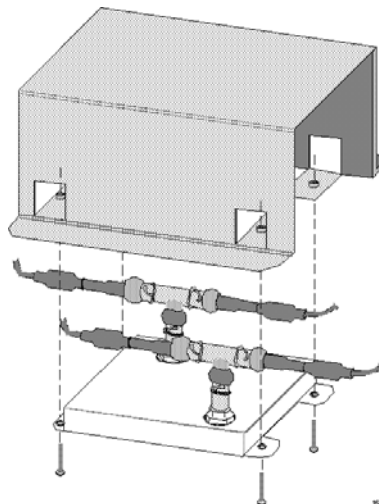
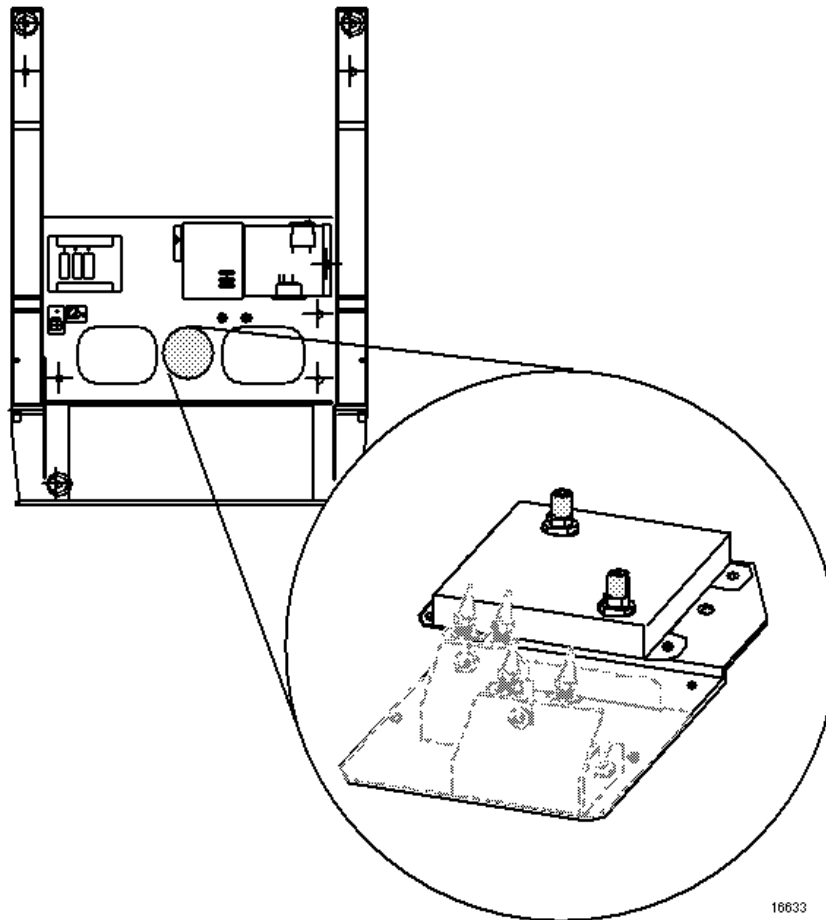


Figure 3-2 LCN MAU to LCN Cable T-Connections

Z-Console connection to the LCN coax cable

The mounting for the LCN MAU is located in the footrest area of the Ergonomic (Z) console.



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Figure 3-3 Location of LCN MAU

3. Workstation Operation

3.2. Network Connections

Connecting LCN coax to LCN MAU in Z-Console

Step	Action
1	Connect the color-coded A cables to the color-coded T-connector on the BNC connector marked A.
2	Connect the color-coded B cables to the color-coded T-connector on the BNC connector marked B.

Classic Console connection to the LCN Coax Cable

The LCN MAU is mounted to the EMC cover that is attached to the rear of the platform enclosure.

Step	Action
1	Follow the color-coding while connecting the color-coded A cables to the color-coded T-connector on the BNC connector marked A.
2	Follow the color-coding while connecting the color-coded B cables to the color-coded T-connector on the BNC connector marked B.

4. Workstation Servicing

4.1 Accessing the Enclosure of the Workstation

Overview

This section contains procedures for removing the workstation from a Z or EZ console, classic console, and a cabinet. The procedures apply to the workstation.

Before you begin servicing



Attention

Perform a complete system shutdown before you begin any of the procedures in this section.



CAUTION

Before you begin any of the procedures in this section, follow the safety instructions in the *Dell System Information Guide*.



Warning

To avoid electrical shock, always unplug your workstation from the electrical outlet before opening the cover.



Attention

Be careful when opening the workstation cover to ensure that you do not accidentally disconnect cables from the system board.

Perform workstation shutdown

The following workstation shutdown procedure is applicable to workstations mounted in a Z or EZ Console, Cassis Console, or Cabinet.

Step	Action
1	Perform a double SHUTDOWN of the native window and a SHUTDOWN of the Windows operating system. Refer to the Global User Station User's

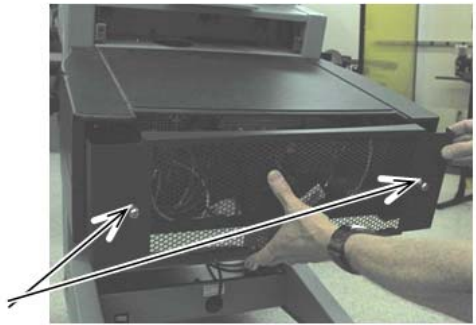
4. Workstation Servicing

4.1. Accessing the Enclosure of the Workstation

Step	Action
	Guide for instructions.
2	When the RESTART message appears on the monitor, place the Power Entry breaker in the OFF position (applicable to consoles only).
3	Turn off power to the computer if it has a separate power switch, and disconnect the AC power cord.
4	Turn off power to the monitor(s) and the printer, if applicable.

Remove the workstation in a Z or EZ Console

If the workstation is mounted in the Z or EZ Console, you will need to remove it from the furniture before servicing.

Step	Action
1	From the rear of the console, loosen the fasteners on the rear cover and remove the rear cover.  Rear Cover fasteners
2	The electronic module sleeve is held in place by detents on the top and bottom of the cover, near the back edge. Lift the back edge of the cover slightly to allow the top detents to clear the top of the workstation. You may also have to push down on the bottom of the cover as you slide it toward you and off the workstation.

4. Workstation Servicing

4.1. Accessing the Enclosure of the Workstation

Step	Action
	
3	Disconnect the cables that are connected to the workstation's back panel connectors. Do not disconnect the peripheral ends of the cables unless specifically told to do so. Tag the cables so that they can be properly reconnected to the workstation.
4	Loosen the M5 hex locknuts on the each retaining bracket and remove them. These will be re-installed later in this document. 
5	Lift the workstation off the frame and place it on a suitable work surface.

4. Workstation Servicing

4.1. Accessing the Enclosure of the Workstation

Remove the workstation from a Classic Console (fixed or slide mount)

If the workstation is mounted using the fixed or slide mount option, you will need to remove it from the furniture before servicing. The pictures below show a fixed mount option.

Step	Action
1	At the front of the console, disconnect the cables that are connected to the workstation's back panel connectors. Do not disconnect the peripheral ends of the cables unless specifically told to do so. Tag the cables so that they can be properly reconnected to the workstation
2	Open the rear console door to access the console enclosure.
3	If the retaining brackets are attached, loosen the two captivated fasteners on each retaining bracket and remove them. These will be re-installed later in this document.  A close-up photograph showing a person's hands using a screwdriver to remove a retaining bracket from the back of a workstation console. The person is wearing a watch on their left wrist. The console is mounted on a stand with wheels. The background is a plain, light-colored wall.

4. Workstation Servicing

4.1. Accessing the Enclosure of the Workstation

Step	Action
4	If using fixed mount option, slide the workstation from the console mounting railings and place it on a suitable work surface. If using slide mount option, pull the slide tray out and place the workstation on a suitable work surface or you may service the workstation while still on the slide tray. 

Remove the workstation from a Cabinet (fixed or slide mount)

If the workstation is mounted using the fixed or slide mount option, you will need to remove it from the furniture before servicing. The picture below shows a fixed mount option.

Step	Action
1	Open the rear cabinet door.
2	Disconnect the cables that are connected to the workstation's back panel connectors. Do not disconnect the peripheral ends of the cables unless specifically told to do so. Tag the cables so that they can be properly reconnected to the workstation.
3	Open the front cabinet door.

4. Workstation Servicing

4.1. Accessing the Enclosure of the Workstation

Step	Action
4	If the retaining brackets are attached, loosen the two captivated fasteners on each retaining bracket and remove them. These will be re-installed later in this document. 
5	If using a fixed mount option, slide the workstation from the cabinet mounting railings and place it on a suitable work surface. If using a slide mount option, pull the slide tray out and place the workstation on a suitable work surface or you may service the workstation while still on the slide tray.

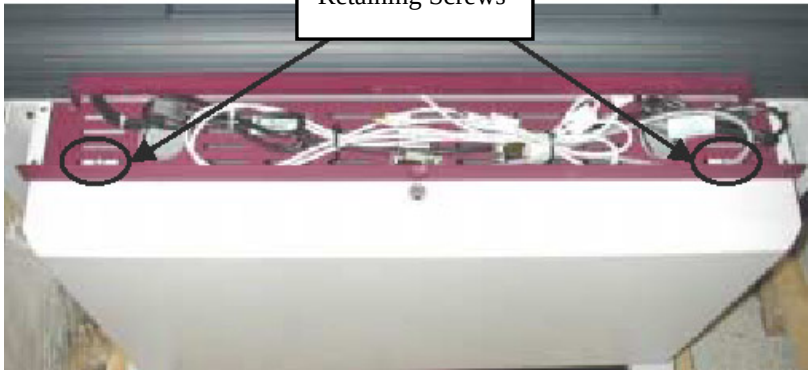
Remove the workstation from a Icon Series Console

If the workstation is mounted in an Icon Series Console, you will need to remove it from the furniture before servicing.

Step	Action
1	Perform a SHUTDOWN of the Windows operating system.
2	Turn off power to the workstation if it has a separate power switch, and disconnect the AC power cord.
3	Turn off power to the console station.

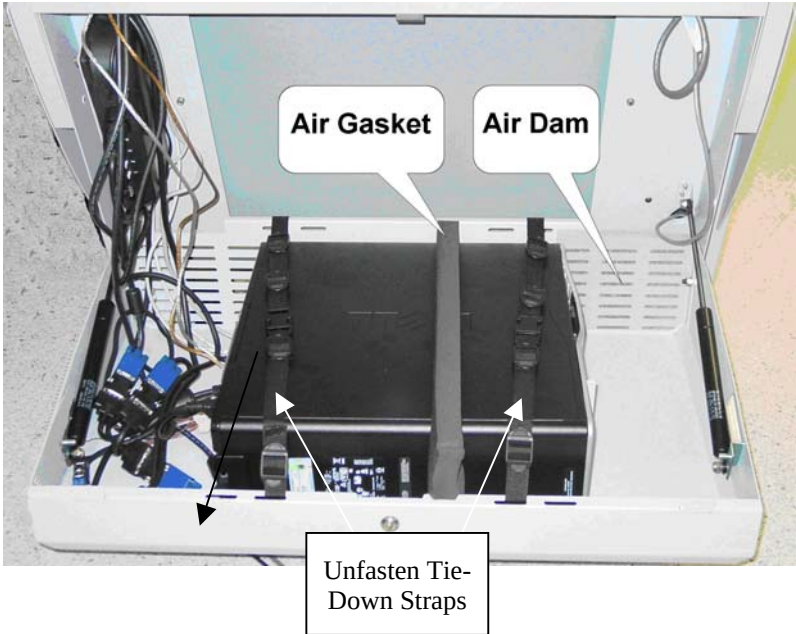
4. Workstation Servicing

4.1. Accessing the Enclosure of the Workstation

Step	Action
4	From the back of the console, remove the two retaining screws that secure the electronic enclosure to the lower valance. 
5	If you have the Power Worksurface version, stand to the side, the enclosure lowers atomically. Use the key to unlock the electronics enclosure and open the latch. If you have the earlier Icon Series Console, hold the top of the electronic enclosure and push down to open it and guide it down to the floor. 
6	Disconnect the cables that are connected to the workstation's back panel connectors. Do not disconnect the peripheral ends of the cables unless specifically told to do so. Tag the cables so that they can be properly reconnected to the workstation.

4. Workstation Servicing

4.1. Accessing the Enclosure of the Workstation

Step	Action
7	Unfasten the tie-down straps and remove air gasket. 
8	Remove the workstation from the enclosure and set on a secure surface.

Open the enclosure

To simplify servicing when using the slide mount option, internal hardware components can be serviced without having to remove the enclosure from the console or cabinet.

Step	Action
1	If this is a slide mount workstation, slide the tray from the furniture

4. Workstation Servicing

4.1. Accessing the Enclosure of the Workstation

Step	Action
2	From the back of the workstation, slide the cover release latch to the left and gently lift up on the cover.  
3	Service the hardware components as required: • For servicing the LCNP4M board, see Section 4.2 in this document. • For servicing other Honeywell installed options, see Section 4.3 in this document. • For other components, see the Dell service manual.

4.2 Servicing the LCNP4M

LCNP4M board description

The LCNP4M board has 16 MB of on-board memory that is soldered to the board.

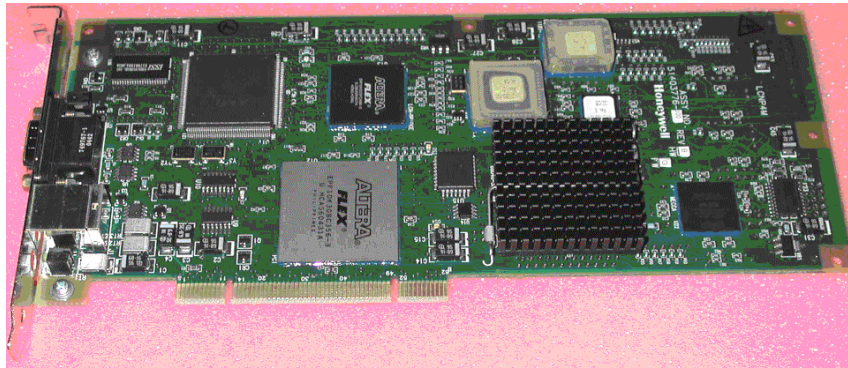


Figure 4-1 LCNP4M Board

Replace the LCNP4M board

The LCNP4M board is located in the PCI-X slot 5 in the workstation. Use this procedure to replace the LCNP4M assembly.

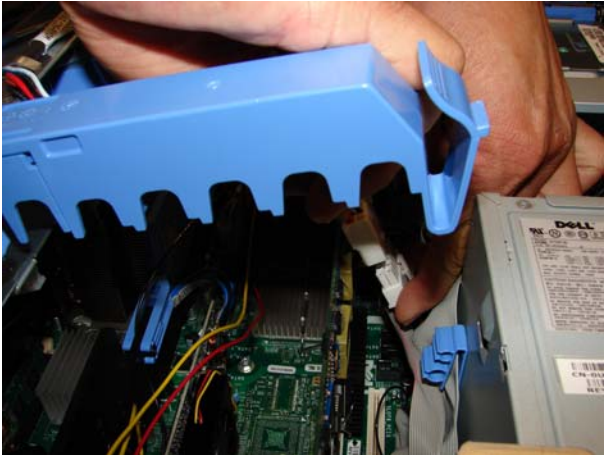


ESD HAZARD

The LCNP4M board is an electrostatically sensitive device. Use a grounding strap and grounded work surfaces and equipment when handling this component. Store and transport parts only in electrostatically safe containers.

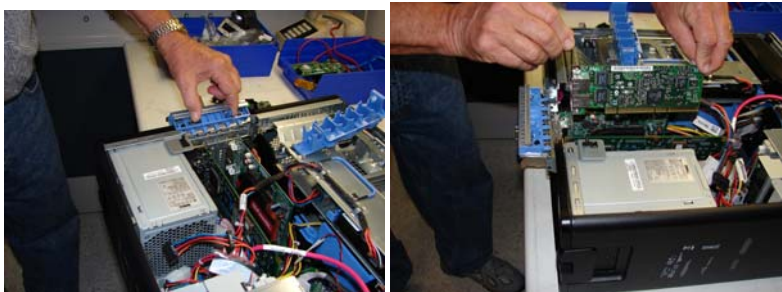
Step	Action
1	Perform all necessary procedures in Section 4.1 to access the LCNP4M assembly including: • Shut down system • Disconnect power from the workstation • Remove the workstation from furniture • Accessing the inside of the electronics enclosure • Disconnect cable from LCNP4M board.

4. Workstation Servicing
4.2. Servicing the LCNP4M

Step	Action
2	Disconnect the LCN MAU cable from the LCNP4M board.
3	Release the tab on the card retention arm, and lift it up to access the LCNP4M board. Disconnect the white connector on the drive power cable.  

4. Workstation Servicing

4.2. Servicing the LCNP4M

Step	Action
4	While wearing a grounded ESD wrist strap, lift the card holder over to expose the mounting screws. Remove the LCNP4M board assembly from the PCI-X slot 5. 
5	Verify that the new LCNP4M board has the correct LCN pin assignments.
6	Insert the new LCNP4M board assembly in the PCI-X slot 5.
7	Reconnect the white connector on the drive power cable. Replace the card holder and card retention arm making sure the tab locks into place.
8	Replace the workstation cover and place the workstation in the furniture using the appropriate installation procedures in Section 2, "Platform Installation".
9	Press the Power On/Off button on the front of the platform to turn the power back on.
10	Run the Configurator to re-establish the correct node number.
11	Reconnect the LCN MAU cable.
12	Restart the Operating System.

4.3 Replacing/Adding Other Options

Overview

This section contains procedures for adding or replacing optional components in the workstation.

Before you begin



CAUTION

Be careful not to damage the EMI gasket fingers when removing/installing boards.



ESD HAZARD

Use a grounding strap and grounded work surfaces and equipment when handling any electrostatically sensitive components such as the video cards, NIC adapter cards, and SCSI controller cards. Store and transport parts only in electrostatically safe containers.

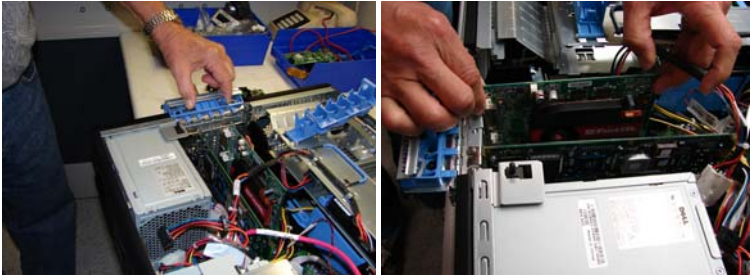
Replace/add video card

The video card is located in the PCI-E slot 2 in the workstation. Use this procedure to replace the video card.

Step	Action
1	Perform all necessary procedures in Section 4.1 to access the video card in the PCI-E slot 2, including: • Shut down system • Disconnect power from the workstation • Remove the workstation from furniture • Accessing the inside of the electronics enclosure • Disconnect cables from the card being replaced.

4. Workstation Servicing

4.3. Replacing/Adding Other Options

Step	Action
2	Remove any adapters and cables from the video card. Release the tab on the card retention arm and lift it up to access the video card slot. Disconnect the white connector on the drive power cable. 
3	While wearing a grounded ESD wrist strap, lift the card holder over to expose the mounting screws. Remove the video assembly from the PCI-E slot 2. 

Step	Action
4	Insert the new single/dual or quad video card in the PCI-E slot 2. Note: Do not mix flat panel displays and CRTs in a multi-screen configuration on a single platform.
5	Reconnect the white connector on the drive power cable. Replace the card holder and card retention arm making sure the tab locks into place.
6	Replace the workstation cover and place the workstation in the furniture using the appropriate installation procedures in Section 2, "Platform Installation".

Replace/add the dual NIC

The dual NIC card is located in the PCI-X slot 6 in the workstation. Use this procedure to replace or add the dual NIC card. If you are adding the FTE dual NIC, you must have access to the *FTE Installation and Service Guide*.

Step	Action
1	Perform all necessary procedures in Section 4.1 to access the dual NIC card in the PCI-X, slot 6, including: • Shut down system • Disconnect power from the workstation • Remove the workstation • Accessing the inside of the electronics enclosure • Disconnect cables to the card being replaced.

4. Workstation Servicing

4.3. Replacing/Adding Other Options

Step	Action
2	Release the tab on the card retention arm, and lift it up to access the NIC card slot. Disconnect the white connector on the drive power cable. 

Step	Action
3	While wearing a grounded ESD wrist strap, lift the card holder over to expose the mounting screws.  If you are replacing an existing NIC, remove it from the PCI-X slot 6.
4	Insert the new NIC card in PCI-X slot 6.
5	Reconnect the white connector on the drive power cable. Replace the card holder and card retention arm making sure the tab locks into place.
6	Replace the workstation cover and place the workstation in the furniture using the appropriate installation procedures in Section 2, "Platform Installation". Note: You will need to reconnect your FTE cables according to the procedures in the <i>FTE Installation and Service Guide</i> in order to verify that the two connectors on the NIC are connected to the right switches.

Replace/add the power adapter card

The power adapter card is located in the PCI-X slot 3 in the workstation. Use this procedure to replace or add the power adapter card.

Step	Action
1	Perform all necessary procedures in Section 4.1 to access the power adapter card in the PCI-X slot 3, including: • Shut down system • Disconnect power from the workstation • Remove the workstation from furniture • Accessing the inside of the electronics enclosure • Disconnect cables to the card being replaced.

4. Workstation Servicing

4.3. Replacing/Adding Other Options

Step	Action
2	Release the tab on the card retention arm and lift it up to access the power adapter card slot. Disconnect the white connector on the drive power cable. 
3	Disconnect the 4-pin Molex connector from the Y cable and remove the power adapter card.
4	Insert the new power adapter card in PCI-X slot 3.
5	Connect the 4-pin molex connector on the power adapter card assembly's cable harness to the Y power cable.
6	Reconnect the white connector on the drive power cable. Replace the card holder and card retention arm making sure the tab locks into place.
7	Replace the workstation cover and place the workstation in the furniture using the appropriate installation procedures in Section 2, "Platform Installation".

Add additional memory to the workstation



ESD HAZARD

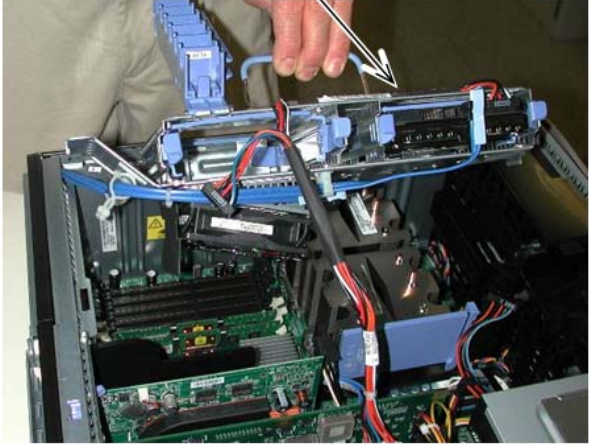
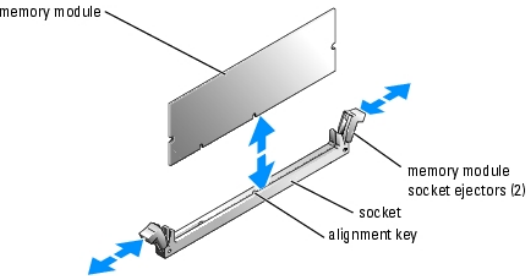
Memory is electrostatically sensitive. Use a grounding strap and grounded work surfaces and equipment when handling these components. Store and transport parts only in electrostatically safe containers.

Use this procedure to add one, two, or three additional gigabits of memory to the workstation. See Figure 1-8 for memory module configuration.

Step	Action
1	Perform all necessary procedures in Section 4.1 to access the memory slots, including: • Shut down system • Disconnect power from the workstation • Remove the workstation from furniture • Accessing the inside of the electronics enclosure.
2	Release the tab on the card retention arm and lift it up. Disconnect the white connector on the drive power cable. 

4. Workstation Servicing

4.3. Replacing/Adding Other Options

Step	Action
3	Lift the card holder by its handle. 
	CAUTION The DIMMs are hot to the touch for some time after the system has been powered down. Allow time for the DIMMs to cool before handling them. Handle the DIMMs by the edges and avoid touching DIMM components.
4	The DIMM sockets are now exposed. Refer to Figure 1-8 for DIMM socket configuration.
5	While wearing a grounded ESD wrist strap, press the ejectors on the memory module socket down and out to allow the memory module to be inserted into the socket. 

Step	Action
6	Align the memory module's connector with the alignment key on the memory module socket, and insert the memory module in the socket. Note: The memory module socket has an alignment key that allows you to install the memory module in the socket in only one way.
7	Press down on the memory module with your thumbs while pulling up on the ejectors with your index fingers to lock the memory module into the socket. When the memory module is properly seated in the socket, the ejectors on the memory module socket align with the ejectors on the other sockets that have memory modules installed.
8	Lower the card holder back into place.
9	Reconnect the white connector on the drive power cable. Replace the card holder and card retention arm making sure the tab locks into place.
10	Replace the workstation cover and place the workstation in the furniture using the appropriate installation procedures in Section 2, "Platform Installation".

4.4 Spare Parts Lists

Ordering workstation spare parts

This section contains lists of spare parts available from Honeywell.

Table 4-1 Spare Parts for the Workstation

Description	Part No.
Mouse USB Optical Mouse	51153747-901
Keyboard USB Keyboard	51153747-902
Expansion RAM 512MB, 667 MHZ, DDR2, ECC, SDRAM, as 1 unit of 512MB DIMM. Must be installed in pairs.	51153731-911
DVD±RW 6, TSST, SATA, BLK	51153747-909
ASSY, DVD+/-RW, 16, Optiarc, SATA, BLK	51153747-910
Floppy Drive 3.5 inch, 1.44 MB	51153747-904
Processor 2.33 GHz Intel® XEON™ Stepped E5410 Processor, 2X6MB shared L2 cache	51153747-911
2.83 GHz Intel® XEON™ Stepped E5440 Processor, 2X6MB shared L2 cache	51153747-912
Video NVIDIA Quadro FX570 Dual Video Card - PEG	51153747-908
Speakers Altec Lansing ADA 215	51199586-911
Software	
Manuals Electronic Documentation	

Other spare parts from Honeywell

The following table lists part numbers for Honeywell furniture.

Table 4-2 Spare Parts for Honeywell Furniture

Description	Part No.
Blank panel front cover, 1U (1.75 in)	51201248-100
Air duct baffle, 1U (1.75 in)	51303521-100
Blank panel front cover, 2U (3.5 in)	51201248-200
Air duct baffle, 2U (3.5 in)	51303521-200
Blank panel front cover, 3U (5.25 in)	51201248-300
Air duct baffle, 3U (5.25 in)	51303521-300
Blank panel front cover, 4U (7 in)	51201248-400
Air duct baffle, 4U (7 in)	51303521-400
Screw, M5 x 16, self tapping for blank front panel covers	51195168-616
Screw, M5 x 12, hex for air baffles	51108888-612
Washer, (8 for front covers, 4 for air baffles)	51109931-100

4. Workstation Servicing

4.5. Verifying Correct BIOS Settings

4.5 Verifying Correct BIOS Settings

Purpose

Honeywell configures specific BIOS settings in the factory for each workstation platform configuration, and this setting should not be altered. BIOS settings for the workstations are listed in **Table 4-3** for workstations with a Non-Raid platform or **Table 4-4** for workstations with a RAID1 platform, so that you may verify the correct settings.

Accessing BIOS

Use this procedure to access BIOS and view the settings. **DO NOT** attempt this procedure unless you are familiar with BIOS.

Step	Action
1	Restart the workstation.
2	When the power on self-test screens appear, press F2 to enter the BIOS Setup.

BIOS Settings for workstations with a Non-Raid platform

This table lists the BIOS settings configured in the factory for the workstation with a non-raid platform.

Table 4-3 BIOS Settings for Workstations with a Non-Raid Platform

System Info	
System	Dell System Workstation T5400
BIOS Version	A04 (08/21/08)
Processor Info CPU	
Type	Intel® XEON™ CPU E5410 at 2.33 GHz or greater
Clock Speed	2.33 GHz or greater
Bus Speed	1333 MHz or greater
L2 Cache	12 MB or greater
ID	0001067A or greater
Family	06
Model	17
Stepping	A
Microcode Version	0000060A
Multi Core Capability	Yes (Quad)
Hyper-threading	No

Capable	
64 Bit	Yes (Intel® EM64T)
Memory Info	
Installed Memory	1.0 GB (Default memory configuration)
Memory Speed	667 MHz
Memory Channel Mode	(Quad) Dual
Memory Technology	FB-DIMM DDR2 SDRAM (ECC)
Boot Sequence	
1. Onboard or USB CD-ROM Drive 2. Onboard or USB Floppy Drive 3. Onboard SATA Hard Drive Onboard Network Controller (not present) Onboard SATA Not Present Onboard USB Device Not Present * Note: Only the devices that are preceded by a number are bootable.	
Drives	
Diskette Drive	Internal
Drive 0: SATA-0	On
Drive 1: SATA-1	Off
Drive 2: SATA-2	On
Drive 3: SATA-3	Off
Drive 4: SATA-4	Off
SATA Operations	RAID Autodetect / AHCI
Smart Reporting	Off
Onboard Devices	
Integrated NIC	Off
Integrated Audio	On
USB Controller	No Boot
Front USB Ports	On
LPT Port Mode	PS/2
LPT Port Address	378H
Serial Port 1	Auto
Serial Port 2	Auto
PS/2 Mouse Port	On
Video	

4. Workstation Servicing

4.5. Verifying Correct BIOS Settings

Primary Video	Option 1
Performance	
SpeedStep	Off
Virtualization	Off
VT for Direct I/O	Off
Limit CPUID Value	Off
HDD Acoustic Mode	Bypass
Snoop Filter	On
ACL Prefetch	Off
HW Prefetch	On
Security	
Admin Password	Not set
System Password	Not set
SATA-0 Password	Not set
SATA-1 Password	This drive is disabled. Drive security is not available.
SATA-2 Password	This drive is a CD or DVD drive. Drive security is not available.
SATA-3 Password	This drive is disabled. Drive security is not available.
Password Changes	Unlocked
Chassis Intrusion	Off
TPM Security	Off
Execute Disable	On
Comptace ®	Deactivate
Power Management	
AC Recovery	Off
Auto Power On	Off
Auto Power Time	12:00 AM
Low Power Mode	Off
Remote Wakeup	Off
Suspend Mode	S1
Maintenance	
Service Tag	XXXXXXX
ASF Mode	Off
SERR Message	Off
Load Defaults	Cancel
Event Log	Mark All Entries

Optional HDD Fan	Not Installed
Post Behavior	
Fast Boot	On
Numlock Key	On
OS Install	Off Note: This BIOS parameter appears when there is 3.0 GB or more of memory installed.
Post Hotkeys	Setup & Boot Menu
Keyboard Errors	Report

BIOS Settings for workstations with a RAID1 platform

This table lists the BIOS settings configured in the factory for the workstation with a RAID1 platform.

Table 4-4 BIOS Settings for Workstation with a RAID1 Platform

System Info	
System	Dell System Workstation T5400
BIOS Version	A04 (08/21/08)
Processor Info CPU	
Type	Intel® XEON™ E5440 CPU 2.83 GHz or greater
Clock Speed	2.83 GHz or greater
Bus Speed	1333 MHz or greater
L2 Cache	12 MB or greater
ID	0001067A or greater
Family	06
Model	17
Stepping	A
Microcode Version	0000060A
Multi Core Capability	Yes (Quad)
Hyper-threading Capable	No
64 Bit	Yes (Intel® EM64T)
Memory Info	
Installed Memory	1.0 GB (Default memory configuration)
Memory Speed	667 MHz
Memory Channel Mode	(Quad) Dual
Memory Technology	FB-DIMM DDR2 SDRAM (ECC)

4. Workstation Servicing

4.5. Verifying Correct BIOS Settings

Boot Sequence	
1. Onboard or USB CD-ROM Drive	
2. Onboard or USB Floppy Drive	
3. Intel Array	
Onboard Network Controller (not present)	
Onboard SATA Not Present	
Onboard USB Device Not Present	
Note: Only devices that are preceded by a number are bootable.	
Drives	
Diskette Drive	Internal
Drive 0: SATA-0	On
Drive 1: SATA-1	On
Drive 2: SATA-2	On
Drive 3: SATA-3	Off
Drive 4: SATA-4	Off
SATA Operations	RAID Autodetect / AHCI
Smart Reporting	Off
Onboard Devices	
Integrated NIC	Off
Integrated Audio	On
USB Controller	No Boot
Front USB Ports	On
LPT Port Mode	PS/2
LPT Port Address	378H
Serial Port 1	Auto
Serial Port 2	Auto
PS/2 Mouse Port	On
Video	
Primary Video	Option 1
Performance	
SpeedStep	Off
Virtualization	Off
VT for Direct I/O	Off
Limit CPUID Value	Off
HDD Acoustic Mode	Bypass

Snoop Filter	On
ACL Prefetch	Off
HW Prefetch	On
Security	
Admin Password	Not set
System Password	Not set
SATA-0 Password	Not set
SATA-1 Password	Not set
SATA-2 Password	This drive is a CD or DVD drive. Drive security is not available.
SATA-3 Password	This drive is disabled. Drive security is not available.
Password Changes	Unlocked
Chassis Intrusion	Off
TPM Security	Off
Execute Disable	On
Comptace ®	Deactivate
Power Management	
AC Recovery	Off
Auto Power On	Off
Auto Power Time	12:00 AM
Low Power Mode	Off
Remote Wakeup	Off
Suspend Mode	S1
Maintenance	
Service Tag	XXXXXXX
ASF Mode	Off
SERR Message	Off
Load Defaults	Cancel
Event Log	Mark All Entries
Optional HDD Fan	Not Installed
Post Behavior	
Fast Boot	On
Numlock Key	On
OS Install	Off Note: This BIOS parameter appears when there is 3.0 GB or more of memory installed.
Post Hotkeys	Setup & Boot Menu
Keyboard Errors	Report

4. Workstation Servicing

4.5. Verifying Correct BIOS Settings

Honeywell

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